Signals and Systems

Detailed Solutions • 1996–2026

GATE INSTRUMENTATION ENGINEERING

(IN)

Himanshu Agarwal & Anish Saha

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Preface

Welcome to PrepFusion™!

This book works, in full, every question GATE has actually asked in Signals and Systems (IN), gathered from the original papers and arranged unit by unit and year by year. Nothing is paraphrased or reconstructed: every question appears as it was set, and every solution is taken step by step rather than asserted.

Each solution carries the same identifier as the question it answers in the companion questions volume, so you can move between practice and explanation without a lookup table. Where the examining authority revised an answer or awarded marks to all (MTA), that is noted with the question it affects.

Used end to end, the book shows you what the paper rewards — which topics repeat, how the framing has shifted over the years, and where your preparation still has gaps.

Meet your Authors

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> *"We built this book the way every aspirant deserves one to be built — organized strictly by unit and by year, so that every hour of practice maps directly onto how GATE actually tests you."*
>
> — Himanshu Agarwal & Anish Saha

How to Read the Question Numbers

Every question in this book carries a three-part identifier instead of a plain serial number:

Q3.24.7

3	Unit (chapter) number — Unit III of this book
24	GATE exam year — 2024 (two digits; 98 = 1998)
7	Question number within that unit and year

So **Q3.24.7** is the 7^{th} question that GATE 2024 contributed to Unit III. Numbers are therefore not sequential through the book — they tell you the unit and the exam year at a glance. The same identifier is used in the questions book, the solutions book and the answer keys, so you can jump between them without a lookup table. In the PDF bookmarks panel, expand a unit to see every question ID and click one to go straight to it.

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Q1.26.1 NAT 2026 2M Ans: 2.5

Given:

$$f(t) = \begin{cases} 1, & t \in [0,2] \\ -t+3, & t \in [2,3] \\ 0, & \text{otherwise} \end{cases}$$

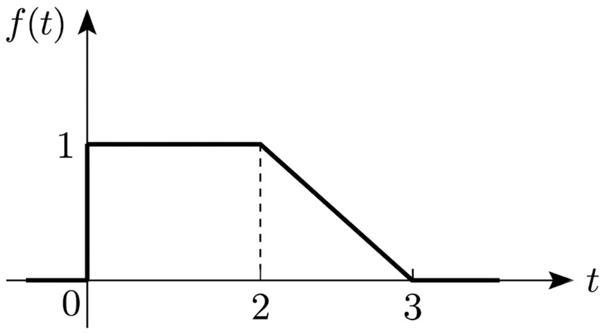

Fig. Solution Q1.26.1

$$\int_{-\infty}^{\infty} f(t)\, dt = 2.5$$

$$\text{Area} = 2 + \frac{1}{2} \times 1 \times 1 = 2.5$$

Hence, the correct answer is **2.5**.

$$\boxed{2.5}$$

Q1.26.2 MCQ 2026 2M Ans: * CLICK FOR VIDEO SOLUTION

We need to determine

$$y(t) = 1 - x(4-t)$$

The transformations are applied in the following order:

(i) $x(t) \to x(t+4)$

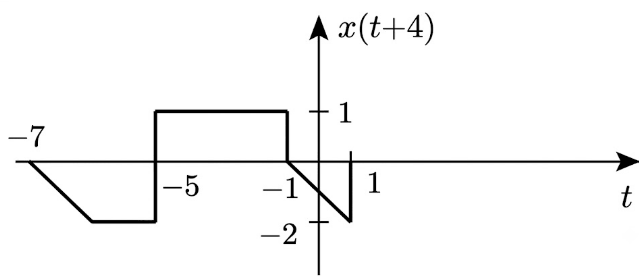

Fig. Solution Q1.26.2

(ii) $x(-t+4)$

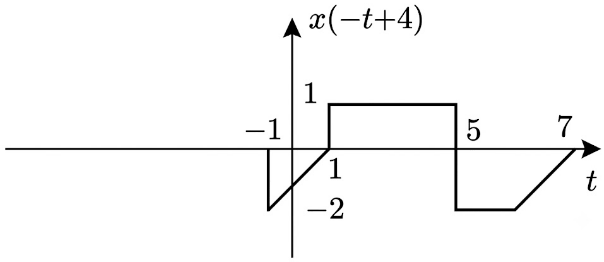

Fig. Solution Q1.26.2

(iii) $y(t) = 1 - x(-t+4)$

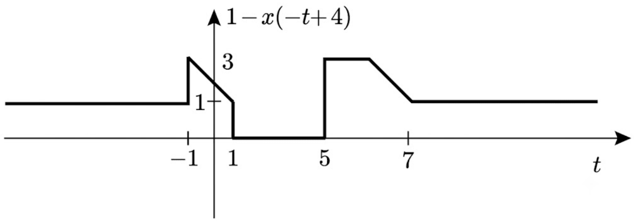

Fig. Solution Q1.26.2

Key Points

- When applying continuous-time transformations to $x(at+b)$, time-shifting ($t \to t+b$) should be performed before time-reversal/scaling.
- Amplitude transformations like $1 - f(t)$ invert the signal vertically across the horizontal axis before adding the DC shift.

Q1.26.3 MCQ 2026 1M Ans: D

Given:

$$x(t) = 4\sin(15\pi t) + 7\cos(4\pi t)$$

For $x_1(t)$, $\omega_1 = 15\pi$,

$$T_1 = \frac{2\pi}{15\pi} = \frac{2}{15}$$

For $x_2(t)$, $\omega_2 = 4\pi$,

$$T_2 = \frac{2\pi}{4\pi} = \frac{1}{2}$$

The sum of periodic signals is periodic if the ratio of their periods is rational.

$$\frac{T_1}{T_2} = \frac{2/15}{1/2} = \frac{4}{15}$$

This is rational, so the signal is periodic.

$$T_0 = \text{LCM}[T_1, T_2]$$

$$T_0 = \text{LCM}\left[\frac{2}{15}, \frac{1}{2}\right]$$

$$T_0 = \frac{\text{LCM}(2,1)}{\text{HCF}(15,2)} = \frac{2}{1}$$

$$T_0 = 2 \text{ seconds}$$

Hence, the correct option is **(D)**.

$$\boxed{\text{(D) } 2}$$

Q1.25.1 MCQ | 2025 | 1M | Ans: B

Given: $x(t) = e^{j9t} + e^{j5t}$ where $j = \sqrt{-1}$,

$$|x(t)| = \left|e^{j9t} + e^{j5t}\right|$$

$$e^{j\theta} = \cos\theta + j\sin\theta$$

$$x(t) = \cos 9t + j\sin 9t + \cos 5t + j\sin 5t$$

$$|x(t)| = \sqrt{(\cos 5t + \cos 9t)^2 + (\sin 5t + \sin 9t)^2}$$

$$|x(t)| = \sqrt{\cos^2 5t + \cos^2 9t + 2\cos 5t\cos 9t + \sin^2 5t + \sin^2 9t + 2\sin 5t\sin 9t}$$

$$|x(t)| = \sqrt{(1 + 1 + 2\cos 5t\cos 9t + 2\sin 5t\sin 9t)}$$

$$|x(t)| = \sqrt{2 + 2\cos 4t}$$

$$|x(t)| = \sqrt{2(1 + \cos 4t)}$$

$$|x(t)| = \sqrt{2(1 + 2\cos^2 2t - 1)}$$

$$|x(t)| = \sqrt{2 \times 2\cos^2 2t}$$

$$|x(t)| = |2\cos 2t|$$

The fundamental period of $\cos(2t)$ is,

$$T = \frac{2\pi}{2} = \pi$$

But for $|\cos(2t)|$, the period is halved.

$$T = \frac{\pi}{2} \text{ sec}$$

$$\boxed{\text{(B) } \frac{\pi}{2}}$$

Q1.23.1 NAT 2023 1M Ans: -3

Given:

$$x(n) = u(-n+5) - u(n+3)$$

The signal $x(n)$ can be plotted as shown below,

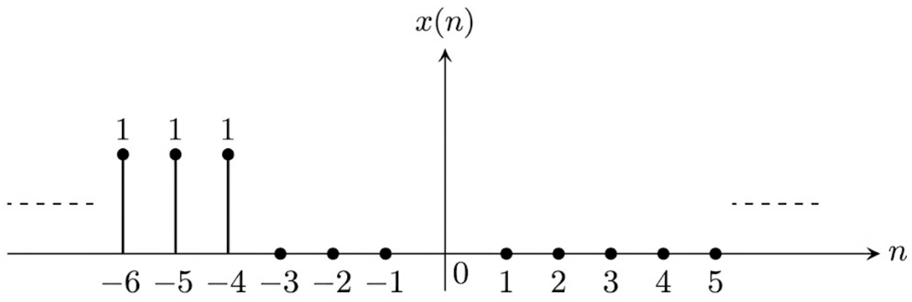

Fig. Solution Q1.23.1

So, the smallest value of n is -3 for $x(n)$ to be zero.
Hence, the correct answer is -3.

$$\boxed{-3.}$$

Q1.23.2 MCQ 2023 1M Ans: C

Theory: For an LTI system, the steady-state response to a sinusoidal input is determined by the magnitude and phase of the frequency response $H(j\omega)$.
The transfer function is

$$H(s) = \frac{s-\pi}{s+\pi}.$$

Since

$$y(t) = \sin(\pi t),$$

the excitation frequency is

$$\omega = \pi \text{ rad/s}.$$

Evaluating the frequency response,

$$H(j\omega) = \frac{j\omega - \pi}{j\omega + \pi}.$$

At $\omega = \pi$,

$$H(j\pi) = \frac{j\pi - \pi}{j\pi + \pi}.$$

The magnitude is

$$|H(j\pi)| = \frac{\sqrt{\pi^2+\pi^2}}{\sqrt{\pi^2+\pi^2}} = 1.$$

The phase is

$$\angle H(j\omega) = 180^\circ - 2\tan^{-1}\left(\frac{\omega}{\pi}\right).$$

Substituting $\omega = \pi$,

$$\angle H(j\pi) = 180^\circ - 2\tan^{-1}(1) = 180^\circ - 90^\circ = 90^\circ.$$

Since

$$Y(j\omega) = H(j\omega)X(j\omega),$$

the output leads the input by 90°. Therefore, the input must lag the output by 90°.
Therefore,

$$\boxed{(C)x(t) = \sin\left(\pi t - \frac{\pi}{2}\right)u(t)}$$

Key Points

- Evaluate the transfer function at the excitation frequency to obtain the steady-state gain and phase.
- The output phase equals the input phase plus the phase of $H(j\omega)$.
- Since $|H(j\pi)| = 1$, only the phase shift affects the input signal.

Mistakes to be avoided

- Do not substitute $s = \pi$; use $s = j\omega$.
- Do not confuse whether the input leads or lags the output; use $Y = HX$.
- Verify both magnitude and phase before selecting the input waveform.

Q1.23.3 NAT 2023 1M Ans: 25

Theory: Time scaling changes the period of a continuous-time periodic signal according to $T' = \dfrac{T}{|a|}$ for $x(at)$.
Given,

$$y(t) = x(4t),$$

and the fundamental period of $x(t)$ is

$$T_x = 100 \text{ s}.$$

If $x(t)$ has fundamental period T, then

$$x(at)$$

has fundamental period

$$T' = \frac{T}{|a|}.$$

Hence,

$$T_y = \frac{100}{4} = 25 \text{ s}.$$

Therefore,

$$\boxed{25}$$

Key Points

- Time compression by a factor of a reduces the period by the same factor.
- For $x(at)$, the new period is $T/|a|$.
- Always use the magnitude $|a|$ while computing the new period.

Mistakes to be avoided

- Do not multiply the period by the scaling factor; divide by $|a|$.
- Do not ignore the absolute value of the time-scaling factor.
- Ensure that the given period is the fundamental period before applying the formula.

Q1.23.4 MCQ | 2023 | 2M | Ans: A | ● ● ○

Theory: For a sinusoidal input, the steady-state output of an LTI system is obtained by evaluating the frequency response $H(j\omega)$ at the input frequency.
Given,

$$h(t) = \delta(t) + 0.5\delta(t-4),$$

and

$$x(t) = \cos\left(\frac{7\pi}{4}t\right).$$

The frequency response is

$$H(j\omega) = 1 + 0.5e^{-j4\omega}.$$

The input frequency is

$$\omega = \frac{7\pi}{4}.$$

Hence,

$$H\left(j\frac{7\pi}{4}\right) = 1 + 0.5e^{-j7\pi} = 1 + 0.5(-1) = 0.5.$$

Since the frequency response is purely real,

$$\angle H\left(j\frac{7\pi}{4}\right) = 0^\circ,$$

there is no phase shift.
Therefore,

$$y(t) = H\left(j\frac{7\pi}{4}\right)x(t) = 0.5\cos\left(\frac{7\pi}{4}t\right).$$

Hence,

$$\boxed{(A) y(t) = 0.5\cos\left(\frac{7\pi}{4}t\right)}$$

Key Points

- The frequency response is the Fourier transform of the impulse response.
- A delayed impulse contributes a factor of $e^{-j\omega\tau}$ in the frequency response.
- For sinusoidal inputs, evaluate $H(j\omega)$ at the input frequency.

Mistakes to be avoided

- Do not substitute $e^{-j7\pi} = 1$; since 7 is odd, $e^{-j7\pi} = -1$.
- Do not perform time-domain convolution when the frequency-response method is simpler.
- Check whether $H(j\omega)$ introduces any phase shift before writing the output.

Q1.21.1 MCQ | 2021 | 1M | Ans: C | ● ○ ○

Given:

$$x(t) = \sin\sqrt{2}\pi t$$

For a non-DC signal to be periodic, there must exist a constant T, such that,

$$x(t \pm T) = x(t) \qquad \cdots (i)$$

Equation (i) can be satisfied only if the power of t in $x(t)$ is 1.

As here the power of t in $x(t)$ is

$$\frac{1}{2}.$$

So, the signal is not periodic.
Hence, the correct option is (C).

$$\boxed{\text{(C) Not periodic}}$$

Q1.20.1 MCQ 2020 1M Ans: B ● ○ ○

Given:

$$x[n] = \sin(2\pi n)u[n],$$

where $u[n] = 1$ for $n \geq 0$.

$$x[0] = \sin 2\pi(0)\, u[0] = 0$$

$$x[1] = \sin 2\pi(1)\, u[1] = 0$$

$$x[2] = \sin 2\pi(2)\, u[2] = 0$$

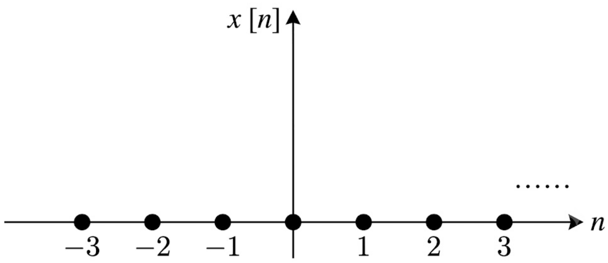

Fig. Solution Q1.20.1

$x[n]$ is a constant (DC) signal of zero value where it repeats its value zero with each value of n. So, the time period = 1.
Hence, the correct option is (B).

$$\boxed{\text{(B) 1}}$$

Q1.20.2 NAT 2020 1M Ans: 0 ● ○ ○

Theory: Convolution with the unit impulse sequence leaves the original sequence unchanged.
Given,

$$g[n] = \delta[n].$$

Hence,

$$y[n] = h[n] \otimes g[n] = h[n] \otimes \delta[n] = h[n].$$

The sequence $h[n]$ is

$$h[n] = \begin{cases} 1, & n = 0, 3, 6, 9, \ldots \\ 0, & \text{otherwise.} \end{cases}$$

Since 2 is not a multiple of 3,

$$h[2] = 0.$$

Therefore,

$$y[2] = h[2] = 0.$$

$$\boxed{\text{y[2]=0}}$$

Key Points

- The sequence $g[n]$ is the unit impulse sequence $\delta[n]$.
- Convolution with $\delta[n]$ satisfies $x[n] \otimes \delta[n] = x[n]$.
- The sequence $h[n]$ is nonzero only at indices $0, 3, 6, \ldots$.

Mistakes to be avoided

- Do not perform lengthy convolution when one sequence is $\delta[n]$.
- Do not confuse convolution with pointwise multiplication.
- Check whether the required index belongs to the set of nonzero indices before evaluating the sequence.

Q1.19.1 NAT 2019 2M Ans: 6

Given:

$$x[n] = e^{j\left(\frac{5\pi}{3}\right)n} + e^{j\left(\frac{\pi}{4}\right)n}$$

For down-sampling by 4,

$$n = 4n$$

$$x[4n] = e^{j\left(\frac{5\pi}{3}\right)4n} + e^{j\left(\frac{\pi}{4}\right)4n}$$

$$\omega_1 = \frac{20\pi}{3}, \qquad \omega_2 = \pi$$

$$N_1 = \frac{2\pi k}{20\pi/3} = \frac{3k}{10}$$

Putting $k = 10$,

$$N_1 = 3$$

$$N_2 = \frac{2\pi k}{\pi} = 2k$$

Putting $k = 1$,

$$N_2 = 2$$

$$(N_1, N_2) = (3, 2)$$

$$\text{Fundamental period} = \text{LCM}(N_1, N_2)$$

$$\text{Fundamental period} = 6$$

Hence, the fundamental period of the down-sampled signal $x_d[n]$ is 6.

$\boxed{6}$

Q1.18.1 MCQ 2018 1M Ans: A

Given:

$$x(t) = \begin{cases} 1, & |t| \le 2 \\ 0, & |t| > 2 \end{cases}$$

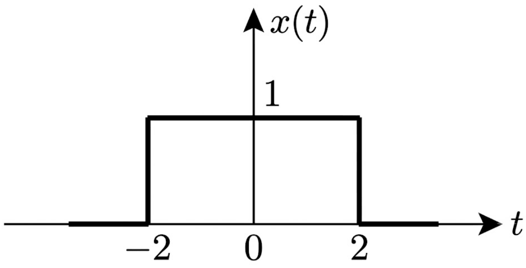

Fig. Solution Q1.18.1

$$I = \int_0^5 2x(t-3)\delta(t-4)\,dt$$

$$I = 2x(4-3)\int_0^5 \delta(t-4)\,dt$$

Since, area under the impulse function is 1,

$$I = 2x(1) \times 1 = 2$$

Hence, the correct option is (A).

$\boxed{\text{(A) } 2}$

Q1.18.2 MCQ 2018 1M Ans: D

Given:

$$z(t) = x(-t) + y(2t+1)$$

where, $x(t)$ and $y(t)$ are periodic signals with period 3 secs.
Period of $x(-t)$ is 3 sec.

$$\text{Period of } y\left[2\left(t+\frac{1}{2}\right)\right] = \frac{3}{2} = 1.5$$

Now, period of $z(t)$ is given by,

$$T = \text{LCM}(3, 1.5) = 3$$

Hence, the correct option is (D).

$\boxed{\text{(D) } 3}$

Key Points

- Time inversion and time shifting do not change the time period of a signal.
- If signal $x(t)$ has period T_0, after time scaling the signal $x(t)$ by factor k, i.e., $x(kt)$,

$$\text{Time period} = \frac{T_0}{k}.$$

Q1.17.1 NAT 2017 1M Ans: 1

Given:

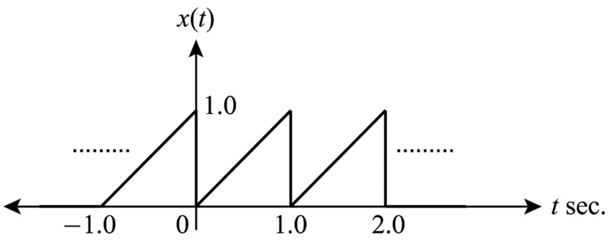

Fig. Solution Q1.17.1

Fundamental period,

$$T_0 = 1 \text{ sec}$$

Fundamental frequency,

$$f_0 = \frac{1}{T_0} = 1 \text{ Hz}$$

Hence, fundamental frequency of signal $x(t)$ is 1 Hz.

$$\boxed{1}$$

Q1.16.1 NAT 2016 2M Ans: 8

Given:

$$x(n) = \sin\left(\frac{301}{4}\pi n\right)$$

where,

$$\omega_0 = \frac{301}{4}\pi \text{ rad/sec}$$

Fundamental time period of discrete-time signal is given by,

$$N_0 = m \times \frac{2\pi}{\omega_0}$$

where, m is the minimum integer number to de-rationalize time period N_0.

$$N_0 = m \times \frac{2\pi}{301\pi/4}$$

$$N_0 = m \times \frac{8}{301}$$

For $m = 301$,

$$N_0 = 8$$

Hence, the fundamental period N_0 is 8.

$$\boxed{8}$$

Q1.15.1 NAT 2015 2M Ans: 6 ● ○ ○

Given:

$$x(t) = 2\cos\left(\frac{2\pi}{3}t\right) + \cos(\pi t)$$

Method Method 1

$$\omega_1 = \frac{2\pi}{3} \text{ rad/sec}, \qquad T_1 = \frac{2\pi}{\omega_1} = \frac{2\pi}{2\pi/3} = 3 \text{ sec}$$

$$\omega_2 = \pi \text{ rad/sec}, \qquad T_2 = \frac{2\pi}{\omega_2} = \frac{2\pi}{\pi} = 2 \text{ sec}$$

Fundamental time period of T_1 and T_2 is calculated as,

	$\times 1$	$\times 2$	$\times 3$
$T_1 = 3$	3	6	9
$T_2 = 2$	2	4	6

Fundamental time period,

$$T_0 = 6 \text{ sec}$$

Hence, fundamental time period of the signal is 6 sec.

Method Method 2

$$\omega_1 = \frac{2\pi}{3} \text{ rad/sec}$$

$$\omega_2 = \pi \text{ rad/sec}$$

Fundamental angular frequency is given by,

$$\omega_0 = \text{HCF}(\omega_1, \omega_2)$$

$$\omega_0 = \text{HCF}\left(\frac{2\pi}{3}, \pi\right) = \frac{\pi}{3}$$

$$T_0 = \frac{2\pi}{\omega_0} = \frac{2\pi}{\pi/3} = 6$$

Hence, fundamental time period of the signal is 6 sec.

$$\boxed{6}$$

Q1.13.1 MCQ 2013 1M Ans: A

Given:

$$v(t) = 30\sin 100t + 10\cos 300t + 6\sin\left(500t + \frac{\pi}{4}\right)$$

$$\omega_1 = 100 \text{ rad/sec}$$

$$\omega_2 = 300 \text{ rad/sec}$$

$$\omega_3 = 500 \text{ rad/sec}$$

Fundamental angular frequency is given by,

$$\omega_0 = \text{HCF}(\omega_1, \omega_2, \omega_3)$$

$$\omega_0 = \text{HCF}(100, 300, 500)$$

$$\omega_0 = 100 \text{ rad/sec}$$

Hence, the correct option is (A).

$$\boxed{\text{(A) } 100}$$

Q1.11.1 MCQ 2011 1M Ans: C

Given:

$$x(n) = \sin \omega_0 n$$

For a discrete-time signal to be periodic, the necessary condition is given by,

$$\frac{m}{N_0} = \frac{\omega_0}{2\pi} = \frac{p}{q} = \text{Ratio of integers.}$$

From the given options, only option (C) satisfies the above condition.
Hence, the correct option is (C).

$$\boxed{\text{(C) } \frac{2\pi}{\omega_0} \text{ to be a ratio of integers}}$$

Q1.11.2 MCQ 2011 1M Ans: A

Given:

$$I = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} t^2 e^{-t^2/2} \delta(1 - 2t)\, dt$$

Method Method 1

$$I = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} t^2 e^{-t^2/2} \times \frac{1}{2}\delta\left(t - \frac{1}{2}\right) dt$$

$$I = \frac{1}{2\sqrt{2\pi}} \int_{-\infty}^{\infty} \left(\frac{1}{2}\right)^2 e^{-(1/2)^2/2} \delta\left(t - \frac{1}{2}\right) dt$$

$$I = \frac{1}{8\sqrt{2\pi}} e^{-1/8} \int_{-\infty}^{\infty} \delta\left(t - \frac{1}{2}\right) dt$$

Since, area under the impulse function is 1,

$$I = \frac{1}{8\sqrt{2\pi}} e^{-1/8}$$

Hence the correct option is (A).

Method Method 2

Using integration property,

$$\int_{-\infty}^{\infty} x(t)\delta(t - t_0)\, dt = x(t_0)$$

$$I = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} t^2 e^{-t^2/2} \delta(1 - 2t)\, dt$$

$$I = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} t^2 e^{-t^2/2} \times \frac{1}{2} \delta\left(t - \frac{1}{2}\right) dt$$

$$I = \frac{1}{2\sqrt{2\pi}} \left(\frac{1}{2}\right)^2 e^{-(1/2)^2/2}$$

$$I = \frac{1}{8\sqrt{2\pi}} e^{-1/8}$$

Hence the correct option is (A).

$$\boxed{\text{(A) } \frac{1}{8\sqrt{2\pi}} e^{-1/8}}$$

Q1.10.1 MCQ | 2010 | 1M | Ans: B

Given:

$$I = \int_{-\infty}^{\infty} \delta\left(t - \frac{\pi}{6}\right) 6 \sin t\, dt$$

Method Method 1

$$I = \int_{-\infty}^{\infty} 6 \sin\left(\frac{\pi}{6}\right) \delta\left(t - \frac{\pi}{6}\right) dt$$

$$I = 6 \times \frac{1}{2} \int_{-\infty}^{\infty} \delta\left(t - \frac{\pi}{6}\right) dt$$

Since, area under the impulse function is 1,

$$I = 3 \times 1 = 3$$

Hence the correct option is (B).

Method Method 2

Using integration property,

$$\int_{-\infty}^{\infty} x(t)\delta(t-t_0)\,dt = x(t_0)$$

$$I = \int_{-\infty}^{\infty} \delta\left(t - \frac{\pi}{6}\right) 6\sin t\,dt$$

$$I = 6\sin t\Big|_{t=\pi/6} = 3$$

Hence, the correct option is (B).

$$\boxed{\text{(B) } 3}$$

Q1.09.1 MCQ | 2009 | 1M | Ans: C | ● ○ ○

Given:

$$x(t) = 2\sin 2\pi t + 3\sin 3\pi t$$

where,

$$\omega_1 = 2\pi \text{ rad/sec}$$

and

$$\omega_2 = 3\pi \text{ rad/sec}$$

Fundamental angular frequency,

$$\omega_0 = \text{HCF}(\omega_1, \omega_2) = \text{HCF}(2\pi, 3\pi)$$

$$\omega_0 = \pi \text{ rad/sec}$$

Fundamental time period,

$$T_0 = \frac{2\pi}{\omega_0} = \frac{2\pi}{\pi} = 2 \text{ sec}$$

Hence, the correct option is (C).

$$\boxed{\text{(C) } 2 \text{ sec}}$$

Q1.08.1 MCQ | 2008 | 1M | Ans: D | ● ○ ○

Given:

$$x(n) = e^{j\left(\frac{5\pi}{6}\right)n}$$

where,

$$\omega_0 = \frac{5\pi}{6} \text{ rad/sec}$$

For a discrete-time signal to be periodic,

$$\frac{m}{N_0} = \frac{\omega_0}{2\pi} = \text{Ratio of integers}$$

$$N_0 = \frac{2\pi}{\omega_0} \times m = \frac{2\pi}{5\pi/6} \times m$$

$$N_0 = \frac{12}{5} \times m$$

where, m is the minimum integer number to de-rationalize the time period N_0.
For $m = 5$,

$$N_0 = 12$$

Hence, the correct option is (D).

$$\boxed{\text{(D) } 12}$$

Q1.07.1 MCQ | 2007 | 2M | Ans: C

Given:

$$x(n) = \left(\frac{1}{3}\right)^n u(n)$$

where,

$$u(n) = \begin{cases} 1, & n \geq 0 \\ 0, & n < 0 \end{cases}$$

and

$$y(n) = x(-n), \qquad -\infty < n < \infty$$

$$\sum_{n=-\infty}^{\infty} y(n) = \sum_{n=-\infty}^{\infty} x(-n)$$

$$\sum_{n=-\infty}^{\infty} y(n) = \sum_{n=-\infty}^{\infty} \left(\frac{1}{3}\right)^{-n} u(-n)$$

$$\sum_{n=-\infty}^{\infty} y(n) = \sum_{n=-\infty}^{0} \left(\frac{1}{3}\right)^{-n}$$

$$\sum_{n=-\infty}^{\infty} y(n) = 1 + \left(\frac{1}{3}\right)^1 + \left(\frac{1}{3}\right)^2 + \cdots$$

$$\sum_{n=-\infty}^{\infty} y(n) = \frac{1}{1 - \frac{1}{3}} = \frac{3}{2}$$

Hence, the correct option is (C).

$$\boxed{(C)\ \frac{3}{2}}$$

Q1.07.2 MCQ 2007 1M Ans: A

Given:

$$x(t) = \left(1 + 0.5\cos 40\pi t\right)\cos 200\pi t$$

$$x(t) = \cos 200\pi t + \frac{0.5}{2}\left[\cos 240\pi t + \cos 160\pi t\right]$$

$$x(t) = \cos 200\pi t + 0.25\cos 240\pi t + 0.25\cos 160\pi t$$

$$x(t) = 0.25\cos 160\pi t + \cos 200\pi t + 0.25\cos 240\pi t$$

where,

$$\omega_1 = 160\pi \text{ rad/sec}$$

$$\omega_2 = 200\pi \text{ rad/sec}$$

$$\omega_3 = 240\pi \text{ rad/sec}$$

Fundamental angular frequency,

$$\omega_0 = \text{HCF}(\omega_1, \omega_2, \omega_3)$$

$$\omega_0 = \text{HCF}(160\pi, 200\pi, 240\pi)$$

$$\omega_0 = 40\pi \text{ rad/sec}$$

Fundamental frequency,

$$f_0 = \frac{\omega_0}{2\pi} = \frac{40\pi}{2\pi} = 20 \text{ Hz}$$

Hence, the correct option is (A).

$$\boxed{\text{(A) } 20}$$

Q1.06.1 MCQ 2006 2M Ans: A

Given:

$$x(t) = \cos(1.2\pi t) + \cos(2\pi t) + \cos(2.8\pi t)$$

$$\omega_1 = 1.2\pi \text{ rad/sec} = \frac{6\pi}{5} \text{ rad/sec}$$

$$\omega_2 = 2\pi \text{ rad/sec}$$

$$\omega_3 = 2.8\pi \text{ rad/sec} = \frac{14\pi}{5} \text{ rad/sec}$$

Fundamental angular frequency is given by,

$$\omega_0 = \text{HCF}(\omega_1, \omega_2, \omega_3)$$

$$\omega_0 = \text{HCF}\left(\frac{6\pi}{5}, \frac{2\pi}{1}, \frac{14\pi}{5}\right)$$

$$\omega_0 = \frac{\text{HCF}[6\pi, 2\pi, 14\pi]}{\text{LCM}[5, 1, 5]}$$

$$\omega_0 = \frac{2\pi}{5} \text{ rad/sec}$$

$$f_0 = \frac{\omega_0}{2\pi} = \frac{\frac{2\pi}{5}}{2\pi} = \frac{1}{5} = 0.2 \text{ Hz}$$

Hence, the correct option is (A).

$$\boxed{\text{(A) } 0.2 \text{ Hz}}$$

Q1.06.2 MCQ 2006 1M Ans: C

Given: 2 V DC voltage and 4 V peak-to-peak square wave signal are superimposed i.e.

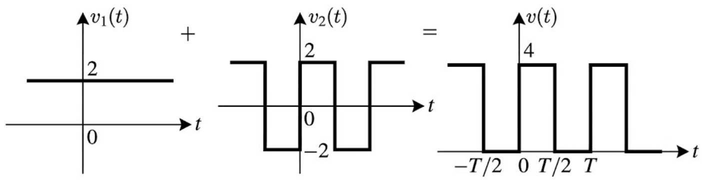

Fig. Solution Q1.06.2

RMS value of superimposed signal is given by,

$$v_{\text{rms}} = \sqrt{\frac{1}{T}\int_{-T/2}^{T/2} v^2(t)\, dt}$$

$$v_{\text{rms}} = \sqrt{\frac{1}{T}\left[\int_{-T/2}^{0} 0\, dt + \int_{0}^{T/2} 4^2\, dt\right]}$$

$$v_{\text{rms}} = \sqrt{\frac{1}{T}\left[16 \times \frac{T}{2}\right]} = \sqrt{8} \text{ V}$$

Hence, the correct option is (C).

$$\boxed{(C)\ \sqrt{8}\text{ V}}$$

Q1.05.1 MCQ 2005 1M Ans: A

Given:

$$x(n) = 3\sin(1.3\pi n + 0.5\pi) + 5\sin(1.2\pi n)$$

$$\omega_1 = 1.3\pi \text{ rad/sec}, \qquad \omega_2 = 1.2\pi \text{ rad/sec}$$

Fundamental angular frequency is given by,

$$\omega_0 = \text{HCF}(\omega_1, \omega_2)$$

$$\omega_0 = \text{HCF}(1.3\pi, 1.2\pi)$$

$$\omega_0 = \text{HCF}\left(\frac{13\pi}{10}, \frac{12\pi}{10}\right)$$

$$\omega_0 = \frac{\text{HCF}[12\pi, 13\pi]}{\text{LCM}[10, 10]}$$

$$\omega_0 = \frac{\pi}{10} \text{ rad/sec}$$

$$T_0 = \frac{2\pi}{\omega_0} = \frac{2\pi}{\pi/10} = 20$$

Hence, the correct option is (A).

$$\boxed{\text{(A) } 20}$$

Key Points

- Fundamental angular frequency:

$$\omega_0 = \text{HCF}(\omega_1, \omega_2, \omega_3, \ldots)$$

- Fundamental period:

$$T = \frac{2\pi}{\omega_0}$$

Q1.01.1 MCQ | 2001 | 2M | Ans: D | ● ○ ○

Power of a discrete-time signal is

$$P = \lim_{N\to\infty} \frac{1}{2N+1} \sum_{n=-N}^{N} |x(n)|^2$$

Option (A): $x(n) = u(n)$

$$P = \lim_{N\to\infty} \frac{1}{2N+1} \sum_{n=0}^{N} 1 = \lim_{N\to\infty} \frac{N+1}{2N+1} = \frac{1}{2}$$

Hence, unit step sequence is a power signal.
Option (B): $x(n) = e^{j\omega_0 n}$

$$P = \lim_{N\to\infty} \frac{1}{2N+1} \sum_{n=-N}^{N} |e^{j\omega_0 n}|^2$$

$$= \lim_{N\to\infty} \frac{1}{2N+1} \sum_{n=-N}^{N} 1 = 1$$

Hence, $e^{j\omega_0 n}$ is a power signal.
Option (C): A periodic sequence.
Every periodic sequence has finite average power. Hence, it is a power signal.

Option (D): $x(n) = nu(n)$

$$P = \lim_{N\to\infty} \frac{1}{2N+1} \sum_{n=0}^{N} n^2$$

$$= \lim_{N\to\infty} \frac{N(N+1)(2N+1)}{6(2N+1)}$$

$$= \lim_{N\to\infty} \frac{N(N+1)}{6} = \infty$$

Thus, the unit ramp sequence is *not* a power signal.

$$\boxed{\text{(D) Unit ramp sequence}}$$

DEBUG INFO

Chapter I

Basics of Signals

Total Questions : 29

MCQ : 21

MSQ : 0

NAT : 8

SOLUTIONS

II

Basics of Systems

Topics

1. Continuous-Time Convolution
2. Discrete-Time Convolution
3. Linear and Non-Linear Systems
4. Time Variant and Time-Invariant Systems
5. LTI Systems - Definition and Properties
6. Causal and Non-Causal Systems
7. Static and Dynamic Systems
8. Stable and Unstable Systems
9. Invertible and Non-Invertible Systems
10. Systems Involving Differential Equations

Q2.26.1 MCQ 2026 2M Ans: A ● ○ ○

(M) $y(t) = x^2(t)$
Linearity:
Let $x(t) = ax_1(t) + bx_2(t)$.

$$y(t) = [ax_1(t) + bx_2(t)]^2$$

$$= a^2x_1^2(t) + b^2x_2^2(t) + 2ab\,x_1(t)x_2(t)$$

Since

$$[ax_1(t) + bx_2(t)]^2 \neq ax_1^2(t) + bx_2^2(t),$$

the system is non-linear.
Time Invariance:
For $x_1(t) = x(t - t_0)$,

$$y_1(t) = [x(t - t_0)]^2$$

and

$$y(t - t_0) = [x(t - t_0)]^2.$$

Hence $y_1(t) = y(t - t_0)$, so the system is time-invariant.

$$(M) \rightarrow (IV)$$

(N) $y(t) = x(-t)$
Linearity:

$$y(t) = ax_1(-t) + bx_2(-t) = ay_1(t) + by_2(t)$$

Hence, the system is linear.
Time Invariance:
For $x_1(t) = x(t - t_0)$,

$$y_1(t) = x(-t - t_0)$$

while

$$y(t - t_0) = x(-t + t_0).$$

Since

$$x(-t - t_0) \neq x(-t + t_0),$$

the system is time-varying.

$$(N) \rightarrow (III)$$

(O) $y(t) = |x(3 - t)|$
Linearity:

$$y(t) = |ax_1(3 - t) + bx_2(3 - t)|$$

but

$$|ax_1(3-t) + bx_2(3-t)| \neq a|x_1(3-t)| + b|x_2(3-t)|.$$

Hence, the system is non-linear.
Time Invariance:
For $x_1(t) = x(t-t_0)$,

$$y_1(t) = |x(3-t-t_0)|$$

while

$$y(t-t_0) = |x(3-t+t_0)|.$$

Since

$$|x(3-t-t_0)| \neq |x(3-t+t_0)|,$$

the system is time-varying.

$$(O) \rightarrow (II)$$

(P) $y(t) = x(t-3)$
Linearity:

$$y(t) = ax_1(t-3) + bx_2(t-3) = ay_1(t) + by_2(t)$$

Hence, the system is linear.
Time Invariance:
For $x_1(t) = x(t-t_0)$,

$$y_1(t) = x(t-t_0-3)$$

and

$$y(t-t_0) = x((t-t_0)-3) = x(t-t_0-3).$$

Hence $y_1(t) = y(t-t_0)$, so the system is time-invariant.

$$(P) \rightarrow (I)$$

Therefore,

$$M-IV, \quad N-III, \quad O-II, \quad P-I$$

$\boxed{\text{(A) M-IV, N-III, O-II, P-I}}$

Q2.25.1 MSQ | 2025 | 2M | Ans: A,C | ● ○ ○

For an LTI system, an eigenfunction satisfies

$$T\{x(t)\} = \lambda x(t)$$

where λ is a scalar.
The eigenfunctions of continuous-time LTI systems are complex exponentials of the form

$$x(t) = e^{st}.$$

Option (A):

$$x(t) = e^{j\frac{2\pi}{3}t}$$

This is a complex exponential. Hence, it is an eigenfunction.
Option (B):

$$x(t) = \cos\left(\frac{2\pi}{3}t\right)$$

A cosine is a sum of two complex exponentials and is not an eigenfunction by itself.
Option (C):

$$x(t) = 2^t = e^{(\ln 2)t}$$

This is an exponential function of the form e^{st}. Hence, it is an eigenfunction.
Option (D):

$$x(t) = \sin\left(\frac{2\pi}{3}t\right)$$

A sine is also a combination of complex exponentials and is not an eigenfunction by itself.

$$\boxed{\text{(A), (C)}}$$

Q2.25.2 MCQ | 2025 | 2M | Ans: C

Given,

$$y[n] = \frac{1}{\alpha}y[n-1] + x[n], \qquad \alpha > 1$$

and

$$x[n] = \delta[n-p], \qquad p > 10.$$

Substituting $x[n]$,

$$y[n] = \frac{1}{\alpha}y[n-1] + \delta[n-p]$$

For $n = p$,

$$y[p] = \frac{1}{\alpha}y[p-1] + \delta[0]$$

$$y[p] = \frac{1}{\alpha}y[p-1] + 1$$

For $n = p + 1$,

$$y[p+1] = \frac{1}{\alpha}y[p] + \delta[1]$$

$$y[p+1] = \frac{1}{\alpha}y[p]$$

Substituting $y[p]$,

$$y[p+1] = \frac{1}{\alpha}\left(\frac{1}{\alpha}y[p-1] + 1\right)$$

$$y[p+1] = \frac{1}{\alpha^2}y[p-1] + \frac{1}{\alpha}$$

Since the system is initially at rest and $p > 10$,

$$y[p-1] = 0$$

Therefore,

$$y[p+1] = \frac{1}{\alpha}$$

$$\boxed{\text{(C) } \frac{1}{\alpha}}$$

Q2.22.1 MCQ 2022 1M Ans: D

Given,

$$y(t) = x(t)\sin(2\pi t)$$

Linearity:
For inputs $x_1(t)$ and $x_2(t)$,

$$y_1(t) = x_1(t)\sin(2\pi t), \qquad y_2(t) = x_2(t)\sin(2\pi t)$$

Then,

$$ay_1(t) + by_2(t) = ax_1(t)\sin(2\pi t) + bx_2(t)\sin(2\pi t)$$

$$= \left[ax_1(t) + bx_2(t)\right]\sin(2\pi t)$$

For input

$$x_3(t) = ax_1(t) + bx_2(t),$$

the output is

$$y_3(t) = \left[ax_1(t) + bx_2(t)\right]\sin(2\pi t).$$

Hence,

$$y_3(t) = ay_1(t) + by_2(t)$$

and the system is linear.
Time Invariance:
For delayed input $x(t - t_0)$,

$$y_1(t) = x(t - t_0)\sin(2\pi t)$$

The delayed version of the original output is

$$y(t-t_0) = x(t-t_0)\sin\big(2\pi(t-t_0)\big)$$

Since

$$x(t-t_0)\sin(2\pi t) \neq x(t-t_0)\sin\big(2\pi(t-t_0)\big),$$

we have

$$y_1(t) \neq y(t-t_0).$$

Therefore, the system is time-varying.

$$\boxed{\text{(D) Linear and time-variant}}$$

Q2.19.1 MCQ | 2019 | 1M | Ans: A

Given,

$$y[n] = \alpha y[n-1] + x[n]$$

Taking the z-transform on both sides,

$$Y(z) = \alpha z^{-1} Y(z) + X(z)$$

$$Y(z)\left(1-\alpha z^{-1}\right) = X(z)$$

Hence, the transfer function is

$$H(z) = \frac{Y(z)}{X(z)} = \frac{1}{1-\alpha z^{-1}}$$

The pole is located at

$$z = \alpha.$$

For a causal discrete-time system to be BIBO stable, all poles must lie inside the unit circle. Therefore,

$$|\alpha| < 1.$$

$$\boxed{\text{(A) } |\alpha| < 1}$$

Q2.14.1 MCQ | 2014 | 1M | Ans: A

Given:

$$h(n) = \begin{cases} \dfrac{\omega_c}{\pi}, & n = 0 \\ \dfrac{\sin(\omega_c n)}{\pi n}, & n \neq 0 \end{cases}$$

Condition for causal system:
Impulse response of the system should satisfy

$$h(n) = 0, \qquad n < 0$$

Since,

$$h(n) = \frac{\sin(\omega_c n)}{\pi n}, \qquad n \neq 0$$

the impulse response is non-zero for $n < 0$.
Therefore, the given system is a non-causal system.

$$h(n) = \frac{\sin(\omega_c n)}{\pi n}$$

Discrete-time Fourier transform of $h(n)$ is shown below.

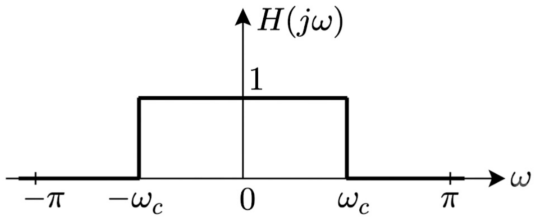

Fig. Solution Q2.14.1

Above figure represents an ideal low-pass filter.

$$\boxed{\text{(A) Non-causal, low-pass filter}}$$

Q2.13.1 | MCQ | 2013 | 1M | Ans: C

For z-plane, condition for stability for a causal system is that all the poles must lie inside

$$|z| = 1.$$

For s-plane, condition for stability for a causal system is that all the poles must lie in the negative half of s-plane.
Statement 1: All the poles of the system must lie on the left side of the $j\omega$ axis.
From above discussion, we can clearly show that statement 1 is true.
Statement 2: Zeros of the system can lie anywhere in the s-plane.
Since stability of the system depends on the poles, hence zeros can lie anywhere in s-plane for a stable system.
Statement 3: All the poles must lie within $|s| = 1$.
There is no such condition for stability in s-plane.
Therefore, statement 3 is wrong.
Statement 4: All the roots of the characteristic equation must be located on the left side of the $j\omega$ axis.
From above discussion, statement 4 is true.
Hence, the correct option is (C).

$$\boxed{\text{(C) All the poles must lie within } |s| = 1.}$$

Q2.13.2 | MCQ | 2013 | 2M | Ans: B

Given: Impulse response,

$$h(t) = \delta(t-1) + \delta(t-3)$$

$$\frac{d}{dt}(\text{Step response}) = \text{Impulse response}$$

$$\frac{d}{dt}[s(t)] = h(t)$$

Thus,

$$s(t) = \int_{-\infty}^{t} h(t)\,dt$$

$$s(t) = \int_{-\infty}^{t} \left[\delta(t-1) + \delta(t-3)\right] dt$$

$$s(t) = u(t-1) + u(t-3)$$

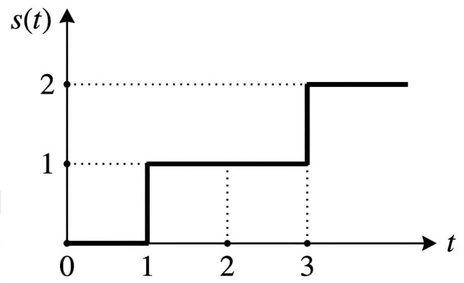

Fig. Solution Q2.13.2

$$s(t)\Big|_{t=2} = 1$$

Hence, the correct option is (B).

$$\boxed{\text{(B) } 1}$$

Q2.12.1 MCQ 2012 2M Ans: D

Given,

$$y(t) = \int_{-\infty}^{t} x(\tau)\cos(3\tau)\,d\tau$$

(i) Time invariance:
Let the input be delayed by t_0. Then,

$$y(t, t_0) = \int_{-\infty}^{t} x(\tau - t_0)\cos(3\tau)\,d\tau$$

Let

$$\tau - t_0 = \tau_1$$

Then,

$$y(t, t_0) = \int_{-\infty - t_0}^{t - t_0} x(\tau_1)\cos\left[3(t_0 + \tau_1)\right] d\tau_1 \tag{i}$$

Replacing t by $t - t_0$,

$$y(t-t_0) = \int_{-\infty}^{t-t_0} x(\tau)\cos(3\tau)\,d\tau \qquad \text{(ii)}$$

From (i) and (ii),

$$y(t,t_0) \neq y(t-t_0)$$

Hence, the system is time-varying.
(ii) Stability:
Let

$$x(t) = \cos(3t)$$

which is a bounded input.
Then,

$$y(t) = \int_{-\infty}^{t} \cos^2(3\tau)\,d\tau$$

Using

$$\cos^2(3\tau) = \frac{1+\cos(6\tau)}{2},$$

$$y(t) = \int_{-\infty}^{t} \frac{1+\cos(6\tau)}{2}\,d\tau$$

$$y(t) = \frac{1}{2}\left[\tau + \frac{\sin(6\tau)}{6}\right]_{-\infty}^{t} = \infty$$

Thus, for a bounded input, the output is unbounded.
Hence, the system is not stable.

$$\boxed{\text{(D) not time-invariant and not stable}}$$

Q2.11.1 MCQ 2011 1M Ans: D

Given,

$$y(t) = \frac{d}{dt}\left\{e^{-t}x(t)\right\}$$

(i) Linearity:
For input $x_1(t)$,

$$y_1(t) = \frac{d}{dt}\left\{e^{-t}x_1(t)\right\}$$

Multiplying by scalar a,

$$a\,y_1(t) = a\frac{d}{dt}\left\{e^{-t}x_1(t)\right\} \qquad \text{(i)}$$

For input $x_2(t)$,

$$y_2(t) = \frac{d}{dt}\left\{e^{-t}x_2(t)\right\}$$

Multiplying by scalar b,

$$b\,y_2(t) = b\frac{d}{dt}\left\{e^{-t}x_2(t)\right\} \tag{ii}$$

Adding (i) and (ii),

$$a\,y_1(t) + b\,y_2(t) = a\frac{d}{dt}\left\{e^{-t}x_1(t)\right\} + b\frac{d}{dt}\left\{e^{-t}x_2(t)\right\} \tag{iii}$$

Now let the input be

$$a\,x_1(t) + b\,x_2(t)$$

with output $y_3(t)$.

$$y_3(t) = \frac{d}{dt}\left\{e^{-t}\big(a\,x_1(t) + b\,x_2(t)\big)\right\}$$

$$y_3(t) = a\frac{d}{dt}\left\{e^{-t}x_1(t)\right\} + b\frac{d}{dt}\left\{e^{-t}x_2(t)\right\} \tag{iv}$$

From (iii) and (iv),

$$y_3(t) = a\,y_1(t) + b\,y_2(t)$$

Therefore, the system satisfies superposition and is ***linear***.

(ii) Time Variance:

For delayed input $x(t-t_0)$,

$$y_1(t) = \frac{d}{dt}\left\{e^{-t}x(t-t_0)\right\} \tag{v}$$

The delayed version of the output is

$$y(t-t_0) = \frac{d}{dt}\left\{e^{-(t-t_0)}x(t-t_0)\right\} \tag{vi}$$

From (v) and (vi),

$$y_1(t) \neq y(t-t_0)$$

Hence, the system is ***time-varying***.

(iii) Stability:

For bounded input

$$x(t) = u(t),$$

$$y(t) = \frac{d}{dt}\left\{e^{-t}u(t)\right\}$$

$$y(t) = e^{-t}\delta(t) + u(t)(-e^{-t})$$

For bounded input, the output is unbounded due to the impulse term.
Therefore, the system is ***unstable***.

(iv) Memory:

A system is memoryless if the output depends only on the present input.
Differentiation depends on neighboring values of the input signal.
Hence, the system possesses ***memory*** and is a dynamic system.

$\boxed{\text{(D) The system has memory}}$

Q2.10.1 MCQ | 2010 | 2M | Ans: C

Given,

$$y(t) = \int_{-\infty}^{5t} x(\tau)\, d\tau$$

(i) Time Variance:
Let the input be delayed by t_0. Then the corresponding output is

$$y_1(t) = \int_{-\infty}^{5t} x(\tau - t_0)\, d\tau \tag{i}$$

Now delay the output $y(t)$ by t_0,

$$y(t - t_0) = \int_{-\infty}^{5(t-t_0)} x(\tau)\, d\tau \tag{ii}$$

From (i) and (ii),

$$y_1(t) \neq y(t - t_0)$$

Therefore, the system is time-varying.
(ii) Causality:
If the output of a system depends only on present and past inputs, then the system is causal. If it depends on future inputs also, then it is non-causal.
Given,

$$y(t) = \int_{-\infty}^{5t} x(\tau)\, d\tau$$

For $t = -1$,

$$y(-1) = \int_{-\infty}^{-5} x(\tau)\, d\tau$$

For $t = 1$,

$$y(1) = \int_{-\infty}^{5} x(\tau)\, d\tau$$

Here, it is clear that the output depends upon both past and future values of the input.
Therefore, the system is non-causal.

$\boxed{\text{(C) time variant and non-causal}}$

Q2.09.1 MCQ | 2009 | 2M | Ans: D

Given,

$$y(t) = \sum_{k=-\infty}^{\infty} x(kT)\, \delta(t - kT)$$

(i) Linearity:
For input $x_1(t)$, output will be $y_1(t)$.

$$y_1(t) = \sum_{k=-\infty}^{\infty} x_1(kT)\, \delta(t - kT)$$

Multiplying both sides by scalar quantity a,

$$a\,y_1(t) = a \sum_{k=-\infty}^{\infty} x_1(kT)\,\delta(t-kT) \tag{i}$$

For input $x_2(t)$, output will be $y_2(t)$.

$$y_2(t) = \sum_{k=-\infty}^{\infty} x_2(kT)\,\delta(t-kT)$$

Multiplying both sides by scalar quantity b,

$$b\,y_2(t) = b \sum_{k=-\infty}^{\infty} x_2(kT)\,\delta(t-kT) \tag{ii}$$

Adding equations (i) and (ii),

$$a\,y_1(t) + b\,y_2(t) = \sum_{k=-\infty}^{\infty} [a\,x_1(kT) + b\,x_2(kT)]\,\delta(t-kT) \tag{iii}$$

Now, let the system have input

$$a\,x_1(t) + b\,x_2(t)$$

and corresponding output be $y_3(t)$.

$$y_3(t) = \sum_{k=-\infty}^{\infty} [a\,x_1(kT) + b\,x_2(kT)]\,\delta(t-kT) \tag{iv}$$

From equations (iii) and (iv),

$$y_3(t) = a\,y_1(t) + b\,y_2(t)$$

Therefore, the system follows the principle of superposition. Hence, the system is linear.
(ii) Time Variance:
In a time invariant system, any delay provided in input must be reflected in output.
Let the input be delayed by t_0, then corresponding output is $y_1(t)$.

$$y_1(t) = \sum_{k=-\infty}^{\infty} x(kT-t_0)\,\delta(t-kT) \tag{v}$$

Now, output $y(t)$ is delayed by t_0,

$$y(t-t_0) = \sum_{k=-\infty}^{\infty} x(kT)\,\delta(t-t_0-kT) \tag{vi}$$

From equations (v) and (vi),

$$y_1(t) \neq y(t-t_0)$$

Thus, the system is time varying.

$$\boxed{\text{(D) linear and time-varying}}$$

Q2.08.1 MCQ 2008 1M Ans: D

Time invariance:
In a time invariant system, any delay provided in input must be reflected in output.
(i) From option (A): $y(n) = n\,x(n)$
Let input be delayed by n_0. Then corresponding output is

$$y_1(n) = n\,x(n - n_0) \tag{i}$$

Now, output $y(n)$ is delayed by n_0,

$$y(n - n_0) = (n - n_0)x(n - n_0) \tag{ii}$$

From equations (i) and (ii),

$$y_1(n) \neq y(n - n_0)$$

Thus, the system is time varying.
(ii) From option (B): $y(n) = x(3n)$
Let input be delayed by n_0. Then corresponding output is

$$y_1(n) = x(3n - n_0) \tag{iii}$$

Now, output $y(n)$ is delayed by n_0,

$$y(n - n_0) = x\big[3(n - n_0)\big] \tag{iv}$$

From equations (iii) and (iv),

$$y_1(n) \neq y(n - n_0)$$

Thus, the system is time varying.
(iii) From option (C): $y(n) = x(-n)$
Let input be delayed by n_0. Then corresponding output is

$$y_1(n) = x(-n - n_0) \tag{v}$$

Now, output $y(n)$ is delayed by n_0,

$$y(n - n_0) = x\big[-(n - n_0)\big] \tag{vi}$$

From equations (v) and (vi),

$$y_1(n) \neq y(n - n_0)$$

Thus, the system is time varying.
(iv) From option (D): $y(n) = x(n - 3)$
Let input be delayed by n_0. Then corresponding output is

$$y_1(n) = x(n - n_0 - 3) \tag{vii}$$

Now, output $y(n)$ is delayed by n_0,

$$y(n - n_0) = x\big[(n - 3) - n_0\big] \tag{viii}$$

From equations (vii) and (viii),

$$y_1(n) = y(n - n_0)$$

Thus, the system is time invariant.

$$\boxed{\text{(D) } y[n] = x[n-3]}$$

Q2.01.1 MCQ 2001 1M Ans: D ● ○ ○

Condition for causality of a LTI system:

$$h(n) = 0; \quad n < 0$$

(i) From option (A): $h(n) = a^n u(n-2)$
This signal will exist only for $n \geq 2$ i.e., for $n < 0$, $h(n) = 0$.
Therefore, it is causal.
(ii) From option (B): $h(n) = a^{n-2} u(n)$
This signal will exist only for $n \geq 0$ i.e., for $n < 0$, $h(n) = 0$.
Therefore, it is causal.
(iii) From option (C): $h(n) = a^{n+2} u(n)$
This signal will exist only for $n \geq 0$ i.e., for $n < 0$, $h(n) = 0$.
Therefore, it is causal.
(iv) From option (D): $h(n) = a^n u(n+2)$
This signal exists only for $n \geq -2$ i.e., for $n < 0$, $h(n) \neq 0$.
Therefore, it is non-causal.
Hence, the correct option is (D).

$$\boxed{\text{(D) } a^n u(n+2)}$$

DEBUG INFO

Chapter II

Basics of Systems

Total Questions : 14

MCQ : 13

MSQ : 1

NAT : 0

SOLUTIONS

III

Continuous Time Fourier Series

Topics

1. Introduction to Fourier Series
2. Exponential Fourier Series
3. Predicting the Nature of a Signal from Fourier Coefficients
4. Power Using Fourier Series Coefficients
5. Properties of Exponential Fourier Series
6. Fourier Series of Periodic Waveforms
7. Trigonometric Fourier Series and Symmetry
8. Polar Form, Dirichlet's Conditions and Gibbs Phenomenon

Q3.11.1 MCQ | 2011 | 1M | Ans: B | CLICK FOR VIDEO SOLUTION

Theory: A periodic signal with half-wave symmetry has zero-valued even harmonics in its exponential Fourier series.
From the waveform,

$$x\left(t+\frac{T}{2}\right) = -x(t),$$

which indicates half-wave symmetry.
The waveform repeats every

$$T = 2.$$

For a signal having half-wave symmetry,

$$a_k = 0, \qquad k = \pm 2, \pm 4, \pm 6, \ldots$$

i.e., all even harmonics are absent.
Hence,

$$\boxed{\text{(B) } a_k = 0 \text{ for even integers and } T = 2}$$

Key Points

- Half-wave symmetry eliminates all even harmonics.
- The fundamental period must be identified before selecting the harmonics.
- Here, the waveform repeats every 2 units, so $T = 2$.

Mistakes to be avoided

- Do not mistake the displayed interval for the fundamental period.
- Half-wave symmetry removes even harmonics, not odd harmonics.
- Always determine the correct value of T before writing the Fourier series coefficients.

Q3.22.1 MCQ | 2022 | 2M | Ans: B | CLICK FOR VIDEO SOLUTION

Theory: The symmetry of a periodic signal determines which Fourier series coefficients are present.
Given,

$$f(x) = \begin{cases} -1 - x, & -1 \le x < 0, \\ 1 - x, & 0 < x \le 1, \end{cases}$$

with period 2.

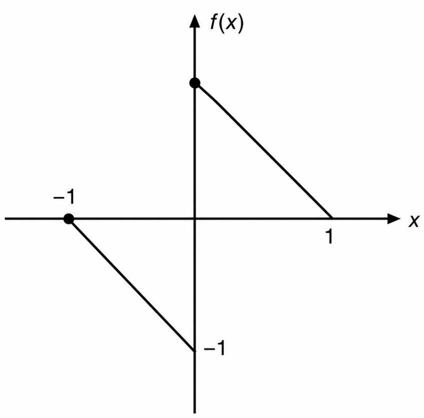

Fig. Solution Q3.22.1

Checking the symmetry,

$$f(-x) = -f(x).$$

Hence, $f(x)$ is an odd function.
For an odd periodic function,

$$a_0 = 0, \qquad a_n = 0,$$

and only the sine coefficients b_n are present.
Since the function does not possess half-wave symmetry, all sine harmonics may exist.
Therefore, the Fourier series contains only

$$\sin(n\pi x), \qquad n = 1, 2, 3, \ldots$$

Hence, the correct option is

$$\boxed{\text{(B) Only } \sin(n\pi x), \text{ where } n = 1, 2, 3, \ldots}$$

Key Points

- Odd functions contain only sine terms in the Fourier series.
- Even functions contain only cosine terms.
- Symmetry should always be checked before evaluating Fourier coefficients.

Mistakes to be avoided

- Do not confuse odd symmetry with half-wave symmetry; they are different properties.
- Do not conclude that only odd harmonics exist unless half-wave symmetry is verified.
- Remember that an odd function has $a_0 = 0$ and $a_n = 0$, leaving only sine coefficients.

Q3.18.1 MCQ 2018 1M Ans: C ● ○ ○ CLICK FOR VIDEO SOLUTION

Theory: An ideal square wave with odd and half-wave symmetry contains only odd harmonics.
The period of the square wave is

$$T = 20 \text{ ms} = 0.02 \text{ s}.$$

Hence, the fundamental frequency is

$$f_0 = \frac{1}{T} = \frac{1}{0.02} = 50 \text{ Hz}.$$

Since the waveform has odd and half-wave symmetry, its spectrum contains only odd harmonics,

$$50,\ 150,\ 250,\ 350, \ldots \text{ Hz}.$$

The ideal LPF has cut-off frequency

$$f_c = 120 \text{ Hz}.$$

Therefore, only the 50 Hz component lies within the passband.
Hence, the output is a pure sinusoid of frequency 50 Hz.

$$\boxed{\text{Option (C): Output is a pure sinusoid of frequency 50 Hz.}}$$

Key Points

- The fundamental frequency is $f_0 = \dfrac{1}{T}$.

- An odd square wave contains only odd harmonics.
- An ideal LPF removes all harmonics above its cut-off frequency.

Mistakes to be avoided

- Do not include the 100 Hz component; it is an even harmonic and is absent.
- Do not assume the output remains a square wave after low-pass filtering.
- Always compare the harmonic frequencies with the LPF cut-off frequency before selecting the output components.

Q3.16.1 MCQ 2016 2M Ans: A ● ○ ○ CLICK FOR VIDEO SOLUTION

Theory: A periodic signal with half-wave symmetry contains only odd harmonics; all even harmonics are absent.
From the waveform,

$$x\left(t+\frac{T}{2}\right) = -x(t),$$

which indicates half-wave symmetry.
Hence,

All even harmonic Fourier coefficients are zero.

Since the 10^{th} harmonic is an even harmonic,

$$C_{10} = 0.$$

Therefore, the power in the 10^{th} harmonic is

$$P_{10} = |C_{10}|^2 = 0.$$

$$\boxed{\text{(A) } 0}$$

Key Points

- Half-wave symmetry eliminates all even harmonics.
- Harmonic power is zero whenever the corresponding Fourier coefficient is zero.
- Only odd harmonics contribute to the Fourier series of this waveform.

Mistakes to be avoided

- Do not compute Fourier coefficients unnecessarily after identifying half-wave symmetry.
- Do not confuse odd symmetry with half-wave symmetry.
- Remember that the 10^{th} harmonic is an even harmonic.

Q3.10.1 MCQ 2010 1M Ans: A ● ○ ○ CLICK FOR VIDEO SOLUTION

Theory: A periodic signal with half-wave symmetry has zero average value over one period.
From the waveform,

$$f\left(x+\frac{T}{2}\right) = -f(x),$$

which indicates half-wave symmetry.

For a trigonometric Fourier series,

$$f(x) = a_0 + \sum_{n=1}^{\infty} (a_n \cos(nx) + b_n \sin(nx)),$$

the constant coefficient a_0 represents the average (DC) value of the signal.
Since the positive and negative halves of one period cancel each other,

$$a_0 = 0.$$

Hence,

$$\boxed{\text{(A) } a_0 = 0}$$

Key Points

- Half-wave symmetry implies $f(x + T/2) = -f(x)$.
- Signals with half-wave symmetry have zero average value.
- Therefore, the DC component a_0 is zero.

Mistakes to be avoided

- Do not confuse half-wave symmetry with even or odd symmetry.
- Remember that a_0 denotes the DC (average) value.
- If half-wave symmetry exists, the average value is always zero.

DEBUG INFO

Chapter III

Continuous Time Fourier Series

Total Questions : 5

MCQ : 5

MSQ : 0

NAT : 0

SOLUTIONS

IV

Continuous Time Fourier Transform

Topics

1. Introduction to Fourier Transform
2. Properties of Fourier Transform
3. Fourier Transform of Standard Signals
4. Duality Property
5. FT of Rectangular, Triangular and Sampling Signals
6. Fourier Transform of Periodic Signals
7. Differentiation, Integration, Multiplication and Convolution Property
8. Parseval's Theorem
9. LTI Systems and Analog Filters
10. Nyquist Rate - Baseband and Bandpass Signals
11. Sampling Theorem and Applications

Q4.25.1 MCQ 2025 2M Ans: C ● ● ○ CLICK FOR VIDEO SOLUTION

Theory: The Fourier transform of the cross-correlation of two real-valued signals is equal to the product of the Fourier transform of one signal and the complex conjugate of the other.
Given,

$$h(t) = R_{f,g(-t)}(t).$$

The Fourier transform of the cross-correlation is

$$H(j\omega) = F(j\omega)\left[G_{g(-t)}(j\omega)\right]^*.$$

Using the time-reversal property,

$$g(-t) \overset{\mathcal{F}}{\longleftrightarrow} G(-j\omega).$$

Hence,

$$H(j\omega) = F(j\omega)\,[G(-j\omega)]^*.$$

Since $g(t)$ is real-valued,

$$G^*(\omega) = G(-\omega).$$

Therefore,

$$[G(-j\omega)]^* = G(j\omega),$$

which gives

$$H(j\omega) = F(j\omega)G(-j\omega).$$

Hence, the correct answer is

$$\boxed{\text{(C) } F(j\omega)G(-j\omega)}$$

Key Points

- The Fourier transform of cross-correlation is the product of one spectrum and the complex conjugate of the other.
- Time reversal in the time domain corresponds to frequency reversal in the frequency domain.
- For real-valued signals, $G^*(\omega) = G(-\omega)$.

Mistakes to be avoided

- Do not confuse correlation with convolution.
- Do not forget the complex conjugate in the cross-correlation property.
- Apply the time-reversal property before using the conjugate property for real-valued signals.

Q4.25.2 MCQ 2025 2M Ans: 5 ● ○ ○ CLICK FOR VIDEO SOLUTION

Theory: Sampling the DTFT at N equally spaced frequencies gives the N-point DFT, and the corresponding time-domain sequence is obtained by aliasing the original sequence with period N.
Here,

$$N = 3,$$

and

$$Y[k] = X\left(e^{j\omega_k}\right), \qquad \omega_k = \frac{2\pi k}{3}.$$

The corresponding time-domain sequence is

$$y[n] = \sum_{m=-\infty}^{\infty} x[n+3m].$$

For $n = 0$,

$$y[0] = x[0] + x[3].$$

Substituting the given values,

$$y[0] = 1 + 4 = 5.$$

Hence,

$$\boxed{y[0] = 5}$$

Key Points

- Sampling the DTFT at N equally spaced frequencies yields the N-point DFT.
- The corresponding time-domain sequence is obtained by periodic aliasing with period N.
- The aliasing relation is $y[n] = \sum_{m=-\infty}^{\infty} x[n + mN]$.

Mistakes to be avoided

- Do not compute a 3-point DFT directly from only the first three samples.
- Remember that samples separated by multiples of N are added due to aliasing.
- Use the original sequence values while performing the folding operation.

Q4.24.1 MCQ 2024 1M Ans: 0.70 to 0.80 CLICK FOR VIDEO SOLUTION

Theory: The frequency response of a discrete-time LTI system is obtained by replacing z with $e^{j\omega}$ in its transfer function.
Taking the z-transform of

$$y[n] - 0.5\,y[n-2] = ax[n-4],$$

gives

$$Y(z)\left(1 - 0.5z^{-2}\right) = aX(z)z^{-4}.$$

Hence, the transfer function is

$$H(z) = \frac{Y(z)}{X(z)} = \frac{az^{-4}}{1 - 0.5z^{-2}}.$$

The frequency response is

$$H(e^{j\omega}) = \frac{ae^{-j4\omega}}{1 - 0.5e^{-j2\omega}}.$$

Since

$$|e^{-j4\omega}| = 1,$$

the magnitude response is

$$|H(e^{j\omega})| = \frac{|a|}{\left|1 - 0.5e^{-j2\omega}\right|}.$$

At

$$\omega = \frac{\pi}{2},$$

$$e^{-j2\omega} = e^{-j\pi} = -1.$$

Therefore,

$$\left|1 - 0.5e^{-j2\omega}\right| = |1 + 0.5| = 1.5.$$

Thus,

$$\left|H\left(\frac{\pi}{2}\right)\right| = \frac{|a|}{1.5} = 0.5.$$

Hence,

$$|a| = 0.75.$$

Key Points

- Obtain the transfer function using the z-transform of the difference equation.
- The delay term $e^{-j\omega n}$ affects only the phase, not the magnitude.
- Evaluate the magnitude response at the specified frequency to determine the unknown parameter.

Mistakes to be avoided

- Do not include the delay term in magnitude calculations since $|e^{-j\theta}| = 1$.
- Substitute $\omega = \frac{\pi}{2}$ only after deriving the frequency response.
- Evaluate the complex magnitude correctly using $e^{-j\pi} = -1$.

Q4.24.2

MCQ | 2024 | 1M | Ans: 400 | CLICK FOR VIDEO SOLUTION

Theory: The Nyquist sampling frequency is twice the highest frequency component present in the signal.
Using the trigonometric identity,

$$\sin^2\theta = \frac{1-\cos 2\theta}{2},$$

the given signal becomes

$$x(t) = 10\left(\frac{1-\cos(400\pi t)}{2}\right) = 5 - 5\cos(400\pi t).$$

Comparing

$$\cos(400\pi t) = \cos(2\pi f t),$$

gives

$$2\pi f = 400\pi \quad \Rightarrow \quad f = 200 \text{ Hz}.$$

Hence, the highest frequency component is

$$f_{\max} = 200 \text{ Hz}.$$

Therefore, the Nyquist sampling frequency is

$$f_s = 2f_{\max} = 2 \times 200 = 400 \text{ Hz}.$$

Hence,

$$\boxed{400 \text{ Hz.}}$$

Key Points

- Use the identity $\sin^2\theta = \dfrac{1-\cos 2\theta}{2}$ to determine the frequency components.
- The DC component does not affect the Nyquist sampling frequency.
- Nyquist sampling frequency is twice the highest frequency component.

Mistakes to be avoided

- Do not treat 200π as the signal frequency; it is the angular frequency.
- Remember that squaring a sinusoid doubles its frequency.
- Ignore the DC component while determining the highest frequency.

Q4.23.1 MSQ | 2023 | 2M | Ans: B,C | ● ● ○ | CLICK FOR VIDEO SOLUTION

Theory: The Fourier transform of a real even signal is real and even, a real odd signal is imaginary and odd, an imaginary even signal is imaginary and even, and an imaginary odd signal is real and odd.

Consider option (A).
Let
$$y(t) = x(t) + x(-t).$$
Then,
$$y(-t) = x(-t) + x(t) = y(t).$$
Hence, $y(t)$ is a real and even signal.
Therefore, its Fourier transform is real and even, not purely imaginary.
Hence, option (A) is incorrect.

Consider option (B).
Let
$$y(t) = x(t) - x(-t).$$
Then,
$$y(-t) = x(-t) - x(t) = -y(t).$$
Hence, $y(t)$ is a real and odd signal.
Therefore, its Fourier transform is purely imaginary and odd.
Hence, option (B) is correct.

Consider option (C).
Let
$$y(t) = j\left[x(t) + x(-t)\right].$$
Then,
$$y(-t) = j\left[x(-t) + x(t)\right] = y(t).$$
Hence, $y(t)$ is an imaginary and even signal.
Therefore, its Fourier transform is purely imaginary and even.
Hence, option (C) is correct.

Consider option (D).
Let
$$y(t) = j\left[x(t) - x(-t)\right].$$
Then,
$$y(-t) = j\left[x(-t) - x(t)\right] = -y(t).$$
Hence, $y(t)$ is an imaginary and odd signal.
Therefore, its Fourier transform is real and odd, not purely imaginary.
Hence, option (D) is incorrect.

Therefore, the correct options are

$$\boxed{\text{(B) x(t)-x(-t) and (C) j(x(t)+x(-t)}}$$

Key Points

- Real even $\Longleftrightarrow$ Fourier transform is real and even.
- Real odd $\Longleftrightarrow$ Fourier transform is imaginary and odd.
- Imaginary even $\Longleftrightarrow$ Fourier transform is imaginary and even.
- Imaginary odd $\Longleftrightarrow$ Fourier transform is real and odd.

Mistakes to be avoided

- Do not confuse even/odd symmetry with real/imaginary nature of the signal.
- Multiplication by j changes a real signal into an imaginary signal but does not alter its symmetry.
- A purely imaginary Fourier transform can result from either a real odd signal or an imaginary even signal.

Q4.22.1 MSQ 2022 2M Ans: C CLICK FOR VIDEO SOLUTION

Theory: Time scaling changes the bandwidth, whereas time shifting affects only the phase spectrum.
Given,

$$y(t) = x(2t-5) = x\left[2\left(t-\frac{5}{2}\right)\right].$$

The time shift by $\frac{5}{2}$ does not change the magnitude spectrum.
Using the time-scaling property,

$$x(at) \overset{\mathcal{F}}{\longleftrightarrow} \frac{1}{|a|}X\left(\frac{f}{a}\right).$$

Here,

$$a = 2.$$

Hence,

$$Y(f) = \frac{1}{2}X\left(\frac{f}{2}\right)e^{-j2\pi f\left(\frac{5}{2}\right)}.$$

The exponential term contributes only phase and does not affect the bandwidth.

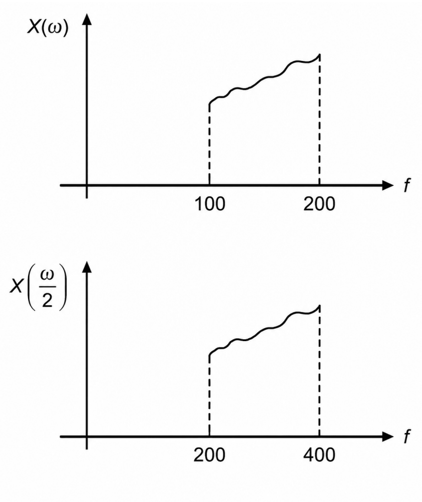

Fig. Solution Q4.22.1

Since $X(f)$ is nonzero for

$$100 \le |f| \le 200 \text{ Hz},$$

the scaled spectrum is nonzero for

$$200 \le |f| \le 400 \text{ Hz}.$$

$$\boxed{\text{(C) y(t) is band-limited between 200Hz and 400Hz}}$$

Mistakes to be avoided

- Do not confuse time scaling with time shifting; only time scaling changes the bandwidth.
- Remember that $x(at)$ scales the frequency axis by the factor a.
- The exponential phase factor due to time shift does not alter the magnitude spectrum.

Q4.20.1 | NAT | 2020 | 1M | Ans: 2 | ● ○ ○ | CLICK FOR VIDEO SOLUTION

Theory: The Fourier transform evaluated at zero frequency equals the total area under the time-domain signal.
Given,

$$x(t) = e^{-|t|}.$$

Using the property,

$$X(j0) = \int_{-\infty}^{\infty} x(t)\,dt.$$

Therefore,

$$\begin{aligned} X(j0) &= \int_{-\infty}^{0} e^{t}\,dt + \int_{0}^{\infty} e^{-t}\,dt \\ &= \left[e^{t}\right]_{-\infty}^{0} + \left[-e^{-t}\right]_{0}^{\infty} \\ &= (1-0) + (1-0) \\ &= 2. \end{aligned}$$

Alternatively, using the known Fourier transform pair,

$$e^{-|t|} \overset{\mathcal{F}}{\longleftrightarrow} \frac{2}{1+\omega^2},$$

and substituting $\omega = 0$,

$$X(j0) = \frac{2}{1+0} = 2.$$

$$\boxed{\text{Answer: } X(j0) = 2}$$

Key Points

- $X(j0)$ equals the area under $x(t)$.
- The signal $e^{-|t|}$ is even, so its area can also be computed as $2\int_0^{\infty} e^{-t}\,dt$.
- The Fourier transform pair $e^{-|t|} \leftrightarrow \dfrac{2}{1+\omega^2}$ is frequently used in GATE problems.

Mistakes to be avoided

- Do not substitute $\omega = 0$ before obtaining the Fourier transform unless using the area property directly.
- Remember that $e^{-|t|} = e^{t}$ for $t < 0$ and e^{-t} for $t \ge 0$.
- Do not forget to split the integral at $t = 0$ when evaluating the area.

Q4.20.2 NAT 2020 2M Ans: 25 ● ● ○ CLICK FOR VIDEO SOLUTION

Theory: Sampling replicates the spectrum at integer multiples of the sampling frequency, while an ideal LPF passes only the frequency components lying within its cut-off frequency.
Given,

$$f_m = 10\text{ kHz}, \qquad f_s = 15\text{ kHz}.$$

After sampling, the spectrum contains replicas at

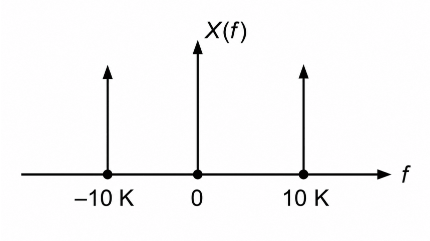

Fig. Solution Q4.20.2

$$f = \pm f_m + nf_s, \qquad n = 0, \pm1, \pm2, \ldots$$

Hence, the frequency components are

$$\pm10,\ \pm5,\ \pm20,\ \pm25,\ \pm35,\ \pm40, \ldots\ \text{kHz}.$$

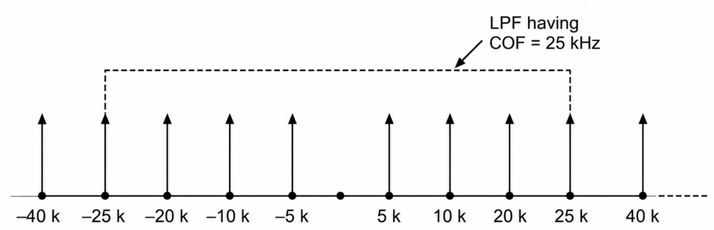

Fig. Solution Q4.20.2

The LPF has cut-off frequency

$$f_c = 25\text{ kHz}.$$

Therefore, the components passed by the LPF are

$$5,\ 10,\ 20,\ 25\text{ kHz}.$$

Hence, the maximum frequency component at the output is

$$25\text{ kHz}.$$

$$\boxed{25\text{ kHz}}$$

Key Points

- Sampling creates spectral replicas at $f = \pm f_m + nf_s$.
- Time shifting does not affect frequency locations; only sampling introduces replicas.
- An ideal LPF passes all frequency components satisfying $|f| \le f_c$.

Mistakes to be avoided

- Do not assume that only the original 10 kHz component exists after sampling.
- Do not confuse the sampling frequency with the LPF cut-off frequency.
- Include the component exactly at the cut-off frequency; 25 kHz is passed by the ideal LPF.

Q4.19.1 NAT 2019 2M Ans: 9 ● ● ○ CLICK FOR VIDEO SOLUTION

Theory: A time delay introduces a phase shift in the Fourier domain without changing the magnitude.
Given,

$$y(t) = x(t) + \frac{1}{2}x(t-1).$$

Taking the Fourier transform,

$$Y(\omega) = X(\omega) + \frac{1}{2}e^{-j\omega}X(\omega) = \left(1 + \frac{1}{2}e^{-j\omega}\right)X(\omega).$$

Hence,

$$|Y(\omega)|^2 = \left|1 + \frac{1}{2}e^{-j\omega}\right|^2 |X(\omega)|^2.$$

At $\omega = 0$,

$$e^{-j0} = 1,$$

therefore,

$$|Y(0)|^2 = \left(1 + \frac{1}{2}\right)^2 |X(0)|^2 = \left(\frac{3}{2}\right)^2 (4).$$

Thus,

$$|Y(0)|^2 = \frac{9}{4} \times 4 = 9.$$

$$\boxed{|Y(0)|^2 = 9}$$

Key Points

- Time delay contributes only a phase factor $e^{-j\omega t_0}$.
- At $\omega = 0$, every delay factor becomes unity.
- Evaluate the transfer function first and then compute the magnitude.

Mistakes to be avoided

- Do not ignore the phase factor before substituting $\omega = 0$.
- Do not add magnitudes directly; first combine the complex terms.
- Remember that $\left(1 + \frac{1}{2}\right)^2 \neq 1 + \frac{1}{4}$.

Q4.18.1 MCQ 2018 2M Ans: C ● ○ ○ CLICK FOR VIDEO SOLUTION

Theory: Multiplication by a complex exponential in the time domain shifts the spectrum in the frequency domain.
Given,

$$y(t) = x(t) + e^{jt}x(t).$$

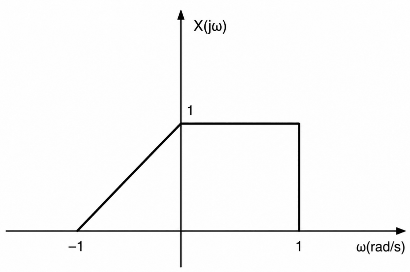

Fig. Solution Q4.18.1

Using the frequency-shift property,

$$x(t)e^{j\omega_0 t} \overset{\mathcal{F}}{\longleftrightarrow} X(\omega - \omega_0),$$

with $\omega_0 = 1$,

$$Y(\omega) = X(\omega) + X(\omega - 1).$$

From the given spectrum,

$$X(0.5) = 1,$$

and

$$X(0.5 - 1) = X(-0.5) = 0.5.$$

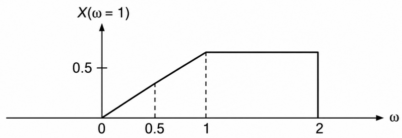

Fig. Solution Q4.18.1

Hence,

$$Y(0.5) = 1 + 0.5 = 1.5.$$

$$\boxed{\text{(C) } 1.5}$$

Key Points

- Multiplication by $e^{j\omega_0 t}$ shifts the spectrum by $+\omega_0$.
- The shifted spectrum is evaluated as $X(\omega - \omega_0)$.
- Read the values directly from the given spectrum after shifting.

Mistakes to be avoided

- Do not use $X(\omega + 1)$ instead of $X(\omega - 1)$.
- Carefully read the amplitude from the shifted graph.
- Remember that spectral shifting changes only the frequency location, not the amplitude.

Q4.17.1 MCQ | 2017 | 1M | Ans: C | ● ○ ○ | CLICK FOR VIDEO SOLUTION

Theory: The sampled sequence is obtained by evaluating the continuous-time signal at the sampling instants.
Given,

$$x(t) = \cos(2\pi t),$$

and

$$f_s = 4 \text{ Hz}.$$

Hence,

$$T_s = \frac{1}{4}.$$

The sampled sequence is

$$x[n] = x(nT_s) = \cos\left(2\pi\frac{n}{4}\right) = \cos\left(\frac{\pi n}{2}\right).$$

At

$$n = 5,$$

$$x[5] = \cos\left(\frac{5\pi}{2}\right) = 0.$$

Therefore,

$$\boxed{\text{(C) } 0}$$

Key Points

- Sample the signal at $t = nT_s$.
- Use $T_s = \frac{1}{f_s}$.
- Evaluate the cosine at the required sample index.

Mistakes to be avoided

- Do not substitute n directly into the continuous-time signal.
- Compute the sampling period before forming the discrete-time sequence.
- Evaluate trigonometric values carefully.

Q4.15.1 MCQ | 2015 | 1M | Ans: C | ● ○ ○ | CLICK FOR VIDEO SOLUTION

Theory: Multiplication in the time domain corresponds to convolution in the frequency domain.
Given,

$$y(t) = x^2(t) = x(t)\,x(t).$$

Therefore,

$$Y(f) = X(f) * X(f).$$

If

$$X(f)$$

is band-limited to

$$f_{\max},$$

then convolution doubles the bandwidth.
Hence,

$$f_{\max,y} = 2f_{\max}.$$

Therefore,

$$\boxed{\text{(C) } 2f_{\max}}$$

Key Points

- Multiplication in time corresponds to convolution in frequency.
- Convolution doubles the maximum frequency support.
- The bandwidth of $x^2(t)$ is twice that of $x(t)$.

Mistakes to be avoided

- Do not confuse squaring the signal with squaring the bandwidth.
- Use the convolution property, not the scaling property.
- The highest frequency doubles, not quadruples.

Q4.12.1 MCQ | 2012 | 2M | Ans: C | ● ○ ○ | CLICK FOR VIDEO SOLUTION

Theory: Multiplication by $\cos(\omega_0\omega)$ in the frequency domain corresponds to averaging two time-shifted versions of the time-domain signal.
Given,

$$H(j\omega) = (2\cos\omega)\,\frac{\sin 2\omega}{\omega}.$$

Using

$$2\cos\omega = e^{j\omega} + e^{-j\omega},$$

we get

$$H(j\omega) = e^{j\omega}\frac{\sin 2\omega}{\omega} + e^{-j\omega}\frac{\sin 2\omega}{\omega}.$$

Let

$$X(j\omega) = \frac{2\sin 2\omega}{\omega}.$$

Then,

$$H(j\omega) = \frac{1}{2}\left[e^{j\omega}X(j\omega) + e^{-j\omega}X(j\omega)\right].$$

Since

$$\frac{2\sin 2\omega}{\omega} \overset{\mathcal{F}}{\longleftrightarrow} x(t) = \begin{cases} 1, & |t| < 2, \\ 0, & \text{otherwise}, \end{cases}$$

using the time-shifting property,

$$h(t) = \frac{1}{2}\left[x(t+1) + x(t-1)\right].$$

Evaluating at $t = 0$,

$$h(0) = \frac{1}{2}\left[x(1) + x(-1)\right].$$

Since

$$x(1) = 1, \qquad x(-1) = 1,$$

we obtain

$$h(0) = \frac{1}{2}(1+1) = 1.$$

$$\boxed{\text{(C) } 1}$$

Key Points

- Multiplication by $e^{\pm j\omega}$ in the frequency domain corresponds to time shifting by ∓ 1 in the time domain.
- $\dfrac{2\sin(\omega T)}{\omega}$ is the Fourier transform of a unit rectangular pulse of width $2T$.
- Evaluate the shifted signals at the required time instant after applying the time-shift property.

Mistakes to be avoided

- Do not forget to rewrite $2\cos\omega$ as $e^{j\omega} + e^{-j\omega}$.
- Do not miss the factor of $\frac{1}{2}$ while expressing $H(j\omega)$ in terms of $X(j\omega)$.
- Carefully apply the time-shift property; the shift direction is often confused.

Q4.11.1 | MCQ | 2011 | 1M | Ans: C | ● ○ ○ | CLICK FOR VIDEO SOLUTION

Theory: The inverse Fourier transform evaluated at $t = 0$ gives the area under the Fourier transform.
Using the inverse Fourier transform,

$$x(t) = \frac{1}{2\pi}\int_{-\infty}^{\infty} X(\omega)e^{j\omega t}\, d\omega.$$

Setting

$$t = 0,$$

we obtain

$$x(0) = \frac{1}{2\pi}\int_{-\infty}^{\infty} X(\omega)\, d\omega.$$

Since

$$x(0) = e^{0} = 1,$$

it follows that

$$\frac{1}{2\pi}\int_{-\infty}^{\infty} X(\omega)\, d\omega = 1.$$

Hence,

$$\boxed{\text{(C) } 1}$$

Key Points

- Use the inverse Fourier transform formula.
- Evaluating the inverse transform at $t = 0$ directly gives the required integral.
- For this signal, $x(0) = 1$.

Mistakes to be avoided

- Do not use Parseval's theorem; it is not applicable here.
- Remember the scaling factor $\frac{1}{2\pi}$ in the inverse Fourier transform.
- Substitute $t = 0$ before evaluating the integral.

Q4.11.2 MCQ 2011 1M Ans: B CLICK FOR VIDEO SOLUTION

Theory: Sampling creates shifted replicas of the spectrum at multiples of the sampling frequency. Components above the Nyquist frequency alias into the baseband.
The given signal is

$$x(t) = \cos(100\pi t) + \sin(300\pi t).$$

The frequency components are

$$50 \text{ Hz} \quad \text{and} \quad 150 \text{ Hz}.$$

Since

$$f_s = 100 \text{ Hz},$$

the Nyquist frequency is

$$f_N = \frac{f_s}{2} = 50 \text{ Hz}.$$

Hence, the 150 Hz component aliases to

$$|150 - 100| = 50 \text{ Hz}.$$

After passing through the ideal LPF of cut-off frequency 100 Hz, the output contains

$$\cos(100\pi t)$$

and the aliased component

$$\sin(100\pi t).$$

Therefore,

$$y(t) \propto \cos(100\pi t) + \sin(100\pi t).$$

$$\boxed{\text{(B) } \cos(100\pi t) + \sin(100\pi t)}$$

Key Points

- Frequencies above the Nyquist frequency alias into the baseband.
- Aliased frequency is $|f - m f_s|$.
- The LPF passes only components within its cut-off frequency.

Mistakes to be avoided

- Do not assume the 150 Hz component is removed completely.
- Compute the aliased frequency before applying the LPF.
- Distinguish between aliasing and filtering.

Q4.08.1 MCQ 2008 1M Ans: C CLICK FOR VIDEO SOLUTION

Theory: The Fourier transform exists only if the signal is absolutely integrable.
Given,

$$x(t) = e^{-at}u(-t).$$

The Fourier transform is

$$X(j\omega) = \int_{-\infty}^{0} e^{-at} e^{-j\omega t}\, dt = \int_{-\infty}^{0} e^{-(a+j\omega)t}\, dt.$$

For the integral to converge,

$$\lim_{t \to -\infty} e^{-(a+j\omega)t} = 0.$$

This requires

$$\mathrm{Re}(a) < 0.$$

Hence, the Fourier transform exists only when the real part of a is strictly negative.

$$\boxed{\text{(C) It exists if the real value of } a \text{ is strictly negative.}}$$

Key Points

- Check the convergence of the exponential at the integration limits.
- For left-sided exponentials, the real part of the exponent must ensure decay as $t \to -\infty$.
- Absolute integrability is the condition for existence of the Fourier transform.

Mistakes to be avoided

- Do not use the convergence condition for $e^{-at}u(t)$, which is different.
- Always consider the support of the unit-step function before applying convergence conditions.
- Do not ignore the sign change caused by the interval $(-\infty, 0]$.

Q4.07.1

MCQ | 2007 | 1M | Ans: C | ● ○ ○ | CLICK FOR VIDEO SOLUTION

Theory: Time expansion compresses the spectrum, whereas time compression expands the spectrum.
The signal

$$x(t)$$

is band-limited to

$$F \text{ Hz}.$$

For

$$x(0.5t),$$

the bandwidth becomes

$$0.5F.$$

Fo

$$x(t),$$

the bandwidth remains

$$F.$$

For

$$x(2t),$$

the bandwidth becomes

$$2F.$$

Hence, the highest frequency component present in

$$y(t) = x(0.5t) + x(t) - x(2t)$$

is

$$2F.$$

Therefore, the Nyquist sampling rate is

$$f_s = 2(2F) = 4F.$$

Hence,

$$\boxed{\text{(C) } 4F}$$

Key Points

- Time scaling changes the bandwidth inversely.
- Time compression increases the bandwidth.
- Nyquist rate is twice the highest frequency present in the signal.

Mistakes to be avoided

- Do not confuse time expansion with bandwidth expansion.
- Determine the maximum bandwidth among all signal components before applying the Nyquist criterion.
- Remember that the Nyquist rate is $2f_{\max}$, not $f_{\max}$.

Q4.06.1 MCQ 2006 2M Ans: D ● ○ ○ CLICK FOR VIDEO SOLUTION

Theory: Compression in the frequency domain corresponds to expansion in the time domain.
From the given spectra,

$$|X(\omega)| \text{ is nonzero over } [-100, 100],$$

whereas

$$|Y(\omega)| \text{ is nonzero over } [-50, 50].$$

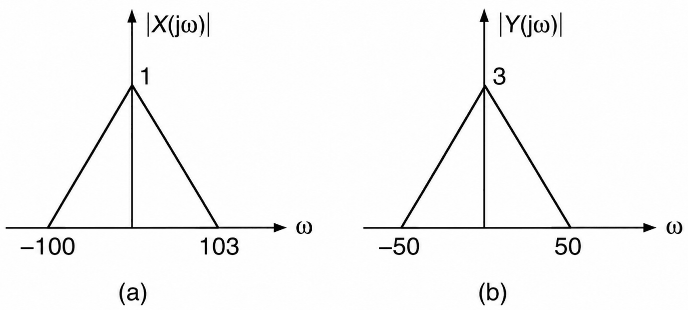

Fig. Solution Q4.06.1

Hence, the spectrum has been compressed by a factor of 2.Using the time-scaling property,

$$x\left(\frac{t}{2}\right) \overset{\mathcal{F}}{\longleftrightarrow} 2X(2\omega).$$

Let

$$y(t) = A\,x\left(\frac{t}{2}\right).$$

Then,

$$Y(\omega) = 2A\,X(2\omega).$$

The peak value of $|X(\omega)|$ is 1, while the peak value of $|Y(\omega)|$ is 3.
Therefore,

$$2A = 3$$

which gives

$$A = \frac{3}{2}.$$

Hence,

$$\boxed{\text{(D) } \frac{3}{2}x\left(\frac{t}{2}\right)}$$

Key Points

- Frequency compression by a factor of 2 corresponds to time expansion by a factor of 2.
- The scaling property is

$$x(at) \overset{\mathcal{F}}{\longleftrightarrow} \frac{1}{|a|}X\left(\frac{\omega}{a}\right).$$

- Always account for the amplitude scaling introduced by time scaling.

Mistakes to be avoided

- Do not forget the amplitude factor associated with time scaling.
- Do not confuse time compression with frequency compression.
- Match both the bandwidth and the peak magnitude before determining the scaling constant.

Q4.06.2

MCQ | 2006 | 2M | Ans: A | CLICK FOR VIDEO SOLUTION

Theory: Express the given spectrum as the sum of standard Fourier transform pairs.
Given,

$$G(\omega) = \frac{\omega^2 + 21}{\omega^2 + 9}.$$

Rewrite it as

$$G(\omega) = 1 + \frac{12}{\omega^2 + 9} = 1 + 2 \cdot \frac{2 \times 3}{\omega^2 + 3^2}.$$

Using the standard Fourier transform pairs,

$$\delta(t) \overset{\mathcal{F}}{\longleftrightarrow} 1,$$

and

$$e^{-a|t|} \overset{\mathcal{F}}{\longleftrightarrow} \frac{2a}{\omega^2 + a^2},$$

with $a = 3$,

$$e^{-3|t|} \overset{\mathcal{F}}{\longleftrightarrow} \frac{6}{\omega^2 + 9}.$$

Therefore,

$$2e^{-3|t|} \overset{\mathcal{F}}{\longleftrightarrow} \frac{12}{\omega^2 + 9}.$$

Hence,

$$g(t) = \delta(t) + 2e^{-3|t|}.$$

$$\boxed{\text{(A) } \delta(t) + 2e^{-3|t|}}$$

Key Points

- Rewrite the spectrum into known transform pairs before taking the inverse transform.
- 1 in the frequency domain corresponds to $\delta(t)$ in the time domain.
- $\dfrac{2a}{\omega^2 + a^2}$ corresponds to $e^{-a|t|}$.

Mistakes to be avoided

- Do not confuse $e^{-a|t|}$ with $e^{-at}u(t)$.
- Always rewrite the rational function before applying transform pairs.
- Remember that $\delta(t)$ appears because of the constant term in the spectrum.

Q4.06.3 MCQ 2006 2M Ans: C

CLICK FOR VIDEO SOLUTION

Theory: Differentiation in the time domain corresponds to multiplication by $(j\omega)^n$ in the frequency domain.
From the given waveform,

$$\frac{d^2 f(t)}{dt^2} = \delta(t-1) - 2\delta(t) + \delta(t+1).$$

Taking the Fourier transform,

$$(j\omega)^2 F(\omega) = e^{-j\omega} - 2 + e^{j\omega}.$$

Using

$$e^{j\omega} + e^{-j\omega} = 2\cos\omega,$$

we get

$$(j\omega)^2 F(\omega) = 2\cos\omega - 2.$$

Hence,

$$F(\omega) = \frac{2\cos\omega - 2}{(j\omega)^2}.$$

Since

$$(j\omega)^2 = -\omega^2,$$

$$F(\omega) = \frac{2(1-\cos\omega)}{\omega^2}.$$

$$\boxed{\text{(C) } \frac{2(1-\cos\omega)}{\omega^2}}$$

Key Points

- $\mathcal{F}\left\{\frac{d^n f(t)}{dt^n}\right\} = (j\omega)^n F(\omega).$
- The Fourier transform of $\delta(t-a)$ is $e^{-j\omega a}$.
- Use Euler's identity to simplify exponential terms.

Mistakes to be avoided

- Do not forget that $(j\omega)^2 = -\omega^2$.
- Carefully apply the sign while dividing by $(j\omega)^2$.
- Remember the shift property of the impulse correctly.

Q4.06.4

MCQ | 2006 | 2M | Ans: C | ● ○ ○ | CLICK FOR VIDEO SOLUTION

Theory: The spacing between adjacent replicas in the sampled spectrum is equal to the sampling frequency.
From the spectrum,

$$f_s = 250 \text{ Hz.}$$

Hence, the sampling interval is

$$T_s = \frac{1}{f_s} = \frac{1}{250} = 0.004 \text{ s.}$$

Therefore,

$$T_s = 4 \text{ ms.}$$

Hence,

$$\boxed{\text{(C) 4 ms}}$$

Key Points

- Adjacent spectral replicas are separated by the sampling frequency.
- Sampling interval and sampling frequency are related by $T_s = \frac{1}{f_s}$.
- Convert seconds into milliseconds carefully.

Mistakes to be avoided

- Do not confuse the signal bandwidth with the sampling frequency.
- Measure the spacing between spectral replicas, not the width of one spectrum.
- Convert 0.004 s correctly to 4 ms.

Q4.06.5

MCQ | 2006 | 2M | Ans: B | ● ○ ○ | CLICK FOR VIDEO SOLUTION

Theory: The output reaches its final value only after all delayed input samples required by the difference equation have become equal to the final input value.
The output is

$$y[n] = \frac{x[n] + x[n-1] + x[n-2] + x[n-4]}{4}.$$

The largest delay involved is

$$4T_s.$$

Since the sampling frequency is

$$f_s = 100 \text{ Hz,}$$

the sampling period is

$$T_s = \frac{1}{f_s} = \frac{1}{100} \text{ s} = 10 \text{ ms.}$$

Hence, the maximum settling time is

$$4T_s = 4 \times 10 = 40 \text{ ms.}$$

Therefore,

$$\boxed{\text{(B) 40 ms}}$$

Key Points

- The maximum delay in the difference equation determines the settling time.
- Compute the sampling period using $T_s = \frac{1}{f_s}$.

- Final output is obtained only after the last delayed sample reaches its final value.

Mistakes to be avoided

- Do not count the number of terms instead of the maximum delay.
- Use the largest delay ($4T_s$), not the average delay.
- Convert seconds to milliseconds correctly.

Q4.05.1 MCQ 2005 1M Ans: A ● ○ ○ CLICK FOR VIDEO SOLUTION

Theory: Multiplication by a cosine in the time domain shifts the spectrum equally to positive and negative frequencies.
Given,

$$x(t) \overset{\mathcal{F}}{\longleftrightarrow} \frac{\pi}{a} e^{-a|\omega|}.$$

Also,

$$\cos(bt) = \frac{e^{jbt} + e^{-jbt}}{2}.$$

Therefore,

$$x(t)\cos(bt) = \frac{1}{2}\left[x(t)e^{jbt} + x(t)e^{-jbt}\right].$$

Using the frequency-shift property,

$$x(t)e^{j\omega_0 t} \overset{\mathcal{F}}{\longleftrightarrow} X(\omega - \omega_0),$$

we obtain

$$\mathcal{F}\{x(t)\cos(bt)\} = \frac{1}{2}\left[X(\omega - b) + X(\omega + b)\right].$$

Substituting

$$X(\omega) = \frac{\pi}{a} e^{-a|\omega|},$$

gives

$$\mathcal{F}\{x(t)\cos(bt)\} = \frac{\pi}{2a}\left[e^{-a|\omega-b|} + e^{-a|\omega+b|}\right].$$

Hence,

$$\boxed{\text{(A) } \frac{\pi}{2a}\left[e^{-a|\omega-b|} + e^{-a|\omega+b|}\right]}$$

Key Points

- Multiplication by $\cos(bt)$ produces two shifted spectra.
- $x(t)e^{j\omega_0 t} \overset{\mathcal{F}}{\longleftrightarrow} X(\omega - \omega_0)$.
- Cosine modulation introduces the average of the positively and negatively shifted spectra.

Mistakes to be avoided

- Do not forget the factor of $\frac{1}{2}$ arising from the cosine expansion.
- Apply the frequency-shift property with the correct sign.
- Preserve the absolute value in the exponential while shifting the spectrum.

Q4.04.1 MCQ | 2004 | 1M | Ans: D | ● ○ ○ | CLICK FOR VIDEO SOLUTION

Theory: The fundamental period of a discrete-time sinusoid is obtained from its normalized angular frequency.
Given,

$$x(t) = 5\cos(150\pi t - 60^\circ),$$

and

$$f_s = 200 \text{ Hz}.$$

Hence,

$$T_s = \frac{1}{f_s} = \frac{1}{200} \text{ s}.$$

The sampled sequence is

$$x[n] = x(nT_s) = 5\cos\left(150\pi\frac{n}{200} - 60^\circ\right).$$

Therefore,

$$x[n] = 5\cos\left(\frac{3\pi}{4}n - 60^\circ\right).$$

Thus, the discrete-time angular frequency is

$$\omega_0 = \frac{3\pi}{4}.$$

Using

$$\omega_0 = \frac{2\pi m}{N},$$

we get

$$\frac{3\pi}{4} = \frac{2\pi m}{N}$$

or

$$\frac{m}{N} = \frac{3}{8}.$$

Since 3 and 8 are coprime,

$$N = 8.$$

Hence,

$$\boxed{\text{(D) } 8}$$

Key Points

- Convert the continuous-time signal into a discrete-time sequence using $T_s = \frac{1}{f_s}$.
- Express the discrete-time frequency as $\omega_0 = \frac{2\pi m}{N}$.
- The fundamental period is the smallest positive integer N.

Mistakes to be avoided

- Do not use the continuous-time period as the discrete-time period.
- Reduce $\frac{m}{N}$ to its lowest terms before determining N.
- The phase angle does not affect the fundamental period.

Q4.03.1 MCQ | 2003 | 1M | Ans: D | ● ○ ○ | CLICK FOR VIDEO SOLUTION

Theory: Time reversal in the time domain corresponds to frequency reversal in the Fourier domain.
Given,

$$f(t) \overset{\mathcal{F}}{\longleftrightarrow} F(\omega).$$

Using the time-reversal property,

$$f(-t) \overset{\mathcal{F}}{\longleftrightarrow} F(-\omega).$$

Since $f(t)$ is real,

$$F(-\omega) = F^*(\omega).$$

Hence,

$$f(t) - f(-t) \overset{\mathcal{F}}{\longleftrightarrow} F(\omega) - F(-\omega) = F(\omega) - F^*(\omega).$$

Now,

$$F(\omega) - F^*(\omega) = 2j\ \text{Im}\{F(\omega)\},$$

which is purely imaginary.
Therefore,

$$\boxed{\text{(D) Imaginary}}$$

Key Points

- $f(-t) \overset{\mathcal{F}}{\longleftrightarrow} F(-\omega)$.
- For a real signal, $F(-\omega) = F^*(\omega)$.
- $F(\omega) - F^*(\omega)$ is always purely imaginary.

Mistakes to be avoided

- Do not confuse frequency reversal with complex conjugation.
- Remember that $F(-\omega) = F^*(\omega)$ holds only for real-valued signals.
- Do not conclude that the result is zero unless $F(\omega)$ itself is purely real.

DEBUG INFO

Chapter IV

Continuous Time Fourier Transform

Total Questions : 25

MCQ : 20

MSQ : 2

NAT : 3

SOLUTIONS

V

Laplace Transform

Topics

1. Introduction to Laplace Transform
2. Laplace Transform of Standard Signals and Properties
3. Region of Convergence (ROC)
4. Inverse Laplace Transform
5. Interrelation Between EFSC, FT and LT
6. Eigen Values and Eigen Functions in LTI Systems
7. Unilateral Laplace Transform

Q5.26.1 MCQ 2026 2M Ans: A ● ● ○ CLICK FOR VIDEO SOLUTION

Theory: Convolution in the time domain corresponds to multiplication in the Laplace domain.
Given,

$$x(t) = u(t-2) - u(t-3),$$

and

$$y(t) = e^{-4t}u(t).$$

The Laplace transform of $x(t)$ is

$$X(s) = \frac{e^{-2s}}{s} - \frac{e^{-3s}}{s} = \frac{e^{-3s}(e^{s}-1)}{s}.$$

Also,

$$Y(s) = \frac{1}{s+4}.$$

Since

$$z(t) = x(t) * y(t),$$

we have

$$Z(s) = X(s)Y(s).$$

Hence,

$$Z(s) = \frac{e^{-3s}(e^{s}-1)}{s(s+4)}.$$

Using

$$e^{-3s}(e^{s}-1) = e^{-2s} - e^{-3s},$$

the expression is already in simplified form.

$$\boxed{\text{(A)}: \frac{e^{-3s}(e^{s}-1)}{s(s+4)}}$$

Key Points

- $\mathcal{L}\{u(t-a)\} = \dfrac{e^{-as}}{s}.$
- Convolution in time becomes multiplication in the Laplace domain.
- Always compute $X(s)$ and $Y(s)$ separately before multiplying.

Mistakes to be avoided

- Do not use the Laplace transform of e^{-4t} without including the unit-step function.
- Do not confuse convolution with multiplication in the time domain.
- While simplifying exponentials, verify identities such as $e^{-3s}(e^{s}-1) = e^{-2s} - e^{-3s}$.

Q5.24.1 MCQ 2024 2M Ans: B ● ● ○ CLICK FOR VIDEO SOLUTION

Theory: For a two-sided Laplace transform, the ROC determines whether each exponential is right-sided or left-sided.
Factorizing the denominator

$$X(s) = \frac{1}{(s+6)(s+7)}.$$

Using partial fractions,

$$\frac{1}{(s+6)(s+7)} = \frac{1}{s+6} - \frac{1}{s+7}.$$

The given ROC is

$$-7 < \Re\{s\} < -6.$$

For

$$\frac{1}{s+6},$$

the ROC must satisfy

$$\Re\{s\} < -6,$$

which corresponds to the left-sided signal

$$-e^{-6t}u(-t).$$

For

$$-\frac{1}{s+7},$$

the ROC must satisfy

$$\Re\{s\} > -7,$$

which corresponds to the right-sided signal

$$-e^{-7t}u(t).$$

Hence, the correct answer is

$$\boxed{\text{(B)}\quad x(t) = -e^{-6t}u(-t) - e^{-7t}u(t)}$$

Key Points

- A right-sided exponential has ROC to the right of its pole.
- A left-sided exponential has ROC to the left of its pole.
- For a two-sided signal, the ROC lies between the poles.

Mistakes to be avoided

- Do not determine the signal without checking the specified ROC.
- Remember that the sign of the residue is retained in the time-domain signal.
- Do not assume both exponentials are right-sided or both are left-sided when the ROC lies between two poles.

Q5.23.1 | MCQ | 2023 | 2M | Ans: C | ● ● ○ | CLICK FOR VIDEO SOLUTION

Theory: Express the signal in delayed form $f(t-T)u(t-T)$ and apply the time-shifting property of the Laplace transform.
Given,

$$x(t) = e^{-3t}u(t-5).$$

Rewrite the signal as

$$x(t) = e^{-3(t-5+5)}u(t-5) = e^{-15}e^{-3(t-5)}u(t-5).$$

Let

$$f(t) = e^{-3t},$$

whose Laplace transform is

$$F(s) = \frac{1}{s+3}, \qquad \text{Re}\{s\} > -3.$$

Using the time-shifting property,

$$\mathcal{L}\{f(t-5)u(t-5)\} = e^{-5s}F(s),$$

we obtain

$$X(s) = e^{-15} \cdot \frac{e^{-5s}}{s+3} = \frac{e^{-5(s+3)}}{s+3}, \qquad \mathrm{Re}\{s\} > -3.$$

Hence,

$$\boxed{(C) X(s) = \frac{e^{-5(s+3)}}{s+3}, \qquad \mathrm{Re}\{s\} > -3}$$

Key Points

- Rewrite the signal in the form $f(t-T)u(t-T)$ before applying the delay property.
- A time delay introduces a multiplicative factor e^{-sT} in the Laplace domain.
- The ROC remains the same as that of the original signal.

Mistakes to be avoided

- Do not treat $e^{-3t}u(t-5)$ as $e^{-3(t-5)}u(t-5)$; include the factor e^{-15}.
- Do not change the ROC because of the time delay.
- Apply the delay property only after expressing the signal as $f(t-T)u(t-T)$.

Q5.23.2

MCQ | 2023 | 2M | Ans: A | CLICK FOR VIDEO SOLUTION

Theory: For an LTI system, if the input is a sinusoid of frequency ω_0, the output is the same frequency scaled by $|H(j\omega_0)|$ and phase shifted by $\angle H(j\omega_0)$.
Given,

$$h(t) = \delta(t) + 0.5\,\delta(t-4).$$

Taking the Fourier transform,

$$H(j\omega) = 1 + 0.5e^{-j4\omega}.$$

The input frequency is

$$\omega_0 = \frac{7\pi}{4}.$$

Therefore,

$$\begin{aligned} H(j\omega_0) &= 1 + 0.5e^{-j4\left(\frac{7\pi}{4}\right)} \\ &= 1 + 0.5e^{-j7\pi} \\ &= 1 - 0.5 \\ &= 0.5. \end{aligned}$$

Hence,

$$|H(j\omega_0)| = 0.5, \qquad \angle H(j\omega_0) = 0.$$

Since

$$x(t) = \cos\left(\frac{7\pi}{4}t\right),$$

the output is

$$\begin{aligned} y(t) &= |H(j\omega_0)| \cos\left(\frac{7\pi}{4}t + \angle H(j\omega_0)\right) \\ &= 0.5\cos\left(\frac{7\pi}{4}t\right). \end{aligned}$$

$$\boxed{\text{(A) } 0.5\cos\left(\frac{7\pi}{4}t\right)}$$

Key Points

- The Fourier transform of $\delta(t-T)$ is $e^{-j\omega T}$.
- An LTI system changes only the amplitude and phase of a sinusoid, not its frequency.
- Evaluate the frequency response only at the input frequency ω_0.

Mistakes to be avoided

- Do not use the Laplace transform; the response to a sinusoidal input is obtained using the frequency response $H(j\omega)$.
- Do not forget to substitute $\omega = \frac{7\pi}{4}$ before evaluating the exponential term.
- Always compute both the magnitude and phase of $H(j\omega_0)$; a negative real value introduces a phase shift of π.

Q5.22.1 | MCQ | 2022 | 1M | Ans: B | CLICK FOR VIDEO SOLUTION

Theory: A time delay in the input introduces the factor e^{-sT} in the Laplace domain, while the transfer function is obtained as $H(s) = \frac{Y(s)}{X(s)}$.

Given,

$$\frac{dy(t)}{dt} + y(t) = 3x(t-3)u(t-3).$$

Taking the Laplace transform assuming zero initial conditions,

$$sY(s) + Y(s) = 3e^{-3s}X(s).$$

Therefore,

$$(s+1)Y(s) = 3e^{-3s}X(s).$$

Hence,

$$H(s) = \frac{Y(s)}{X(s)} = \frac{3e^{-3s}}{s+1}.$$

$$\boxed{(\text{B})\ \frac{3e^{-3s}}{s+1}}$$

Mistakes to be avoided

- Do not forget the exponential delay factor e^{-sT} while taking the Laplace transform of $x(t-T)u(t-T)$.
- Always assume zero initial conditions unless stated otherwise when determining the transfer function.
- Do not confuse the denominator $s+1$ with $s+3$; the coefficient of $y(t)$ determines the denominator.

Q5.22.2 | MCQ | 2022 | 2M | Ans: B | CLICK FOR VIDEO SOLUTION

Theory: A delayed signal $f(t-T)u(t-T)$ has Laplace transform $e^{-sT}F(s)$, where $F(s)$ is the Laplace transform of $f(t)$.
Given,

$$x(t) = (t-1)^2u(t-1).$$

Let

$$f(t) = t^2u(t).$$

Its Laplace transform is

$$F(s) = \frac{2}{s^3}.$$

Since

$$x(t) = f(t-1),$$

using the time-shifting property,

$$X(s) = e^{-s}\frac{2}{s^3}.$$

Evaluating at $s = 1$,

$$X(1) = e^{-1} \cdot \frac{2}{1^3} = \frac{2}{e}.$$

$$\boxed{\text{(B) } \frac{2}{e}}$$

Key Points

- If $f(t) \overset{\mathcal{L}}{\longleftrightarrow} F(s)$, then $f(t-T)u(t-T) \overset{\mathcal{L}}{\longleftrightarrow} e^{-sT}F(s)$.
- $\mathcal{L}\{t^n u(t)\} = \dfrac{n!}{s^{n+1}}$.
- Always identify the undelayed signal before applying the time-shifting property.

Mistakes to be avoided

- Do not directly apply the Laplace transform to $(t-1)^2$; first recognize it as a delayed version of $t^2u(t)$.
- Do not forget the exponential factor e^{-s} introduced by the delay.
- While evaluating $X(1)$, substitute $s = 1$ only after obtaining the complete expression for $X(s)$.

Q5.22.3 MCQ | 2022 | 2M | Ans: B | ● ● ○ | CLICK FOR VIDEO SOLUTION

Theory: For a unity negative-feedback system, the closed-loop transfer function is $T(s) = \dfrac{L(s)}{1+L(s)}$, and stability is determined by the closed-loop poles.
Given,

$$L(s) = \frac{6}{s(s-5)}.$$

The closed-loop transfer function is

$$T(s) = \frac{L(s)}{1+L(s)} = \frac{\dfrac{6}{s(s-5)}}{1+\dfrac{6}{s(s-5)}} = \frac{6}{s(s-5)+6}.$$

Hence,

$$T(s) = \frac{6}{s^2-5s+6} = \frac{6}{(s-2)(s-3)}.$$

The closed-loop poles are

$$s = 2,\ 3.$$

Since both poles lie in the right half of the s-plane, the closed-loop system is **unstable**.
The transfer function is proper since

$$\deg(\text{denominator}) > \deg(\text{numerator}),$$

therefore the system is **causal**.

$$\boxed{\text{(B) Causal and unstable}}$$

Key Points

- For unity negative feedback, $T(s) = \dfrac{L(s)}{1 + L(s)}$.
- Stability is determined by the poles of the closed-loop transfer function.
- A proper transfer function represents a causal LTI system.

Mistakes to be avoided

- Do not judge stability from the open-loop poles; always compute the closed-loop characteristic equation.
- Do not forget to add $1 + L(s)$ while forming the closed-loop transfer function.
- Do not confuse causality with stability; they are independent properties.

Q5.21.1 MCQ | 2021 | 1M | Ans: C | CLICK FOR VIDEO SOLUTION

Theory: A left-sided exponential has a left-half-plane ROC, and the time-reversal property of the bilateral Laplace transform changes s to $-s$.
Given,

$$f(t) = e^t u(-t).$$

We know that

$$e^{-t} u(t) \overset{\mathcal{L}}{\longleftrightarrow} \frac{1}{s+1}, \qquad \Re\{s\} > -1.$$

Using the time-reversal property,

$$x(-t) \overset{\mathcal{L}}{\longleftrightarrow} X(-s),$$

we obtain

$$e^t u(-t) \overset{\mathcal{L}}{\longleftrightarrow} \frac{1}{1-s}, \qquad \Re\{-s\} > -1.$$

Since

$$\frac{1}{1-s} = -\frac{1}{s-1},$$

and

$$\Re\{-s\} > -1 \implies \Re\{s\} < 1,$$

the bilateral Laplace transform is

$$F(s) = -\frac{1}{s-1}, \qquad \Re\{s\} < 1.$$

$$\boxed{\text{(C) } -\frac{1}{s-1} \text{ with real part of } s < 1.}$$

Key Points

- Time reversal changes $X(s)$ to $X(-s)$.
- Left-sided signals have the ROC to the left of the rightmost pole.
- The ROC must also be transformed while applying the time-reversal property.

Mistakes to be avoided

- Do not use the unilateral Laplace transform; the question asks for the bilateral Laplace transform.
- Do not forget to replace s by $-s$ while applying time reversal.
- Always transform the ROC along with the Laplace transform expression.
- Simplify $\frac{1}{1-s}$ carefully as $-\frac{1}{s-1}$.

Q5.21.2 NAT | 2021 | 2M | Ans: 0.33 | ● ○ ○ | CLICK FOR VIDEO SOLUTION

Theory: Convolution in the time domain corresponds to multiplication in the Laplace domain, and a time advance in time corresponds to multiplication by e^{sT} in the Laplace domain.
Given,

$$y(t) = e^{-3t}u(t) * u(t+3).$$

Taking the Laplace transform,

$$\mathcal{L}\{e^{-3t}u(t)\} = \frac{1}{s+3},$$

and

$$\mathcal{L}\{u(t+3)\} = \frac{e^{3s}}{s}.$$

Hence,

$$Y(s) = \frac{1}{s+3} \cdot \frac{e^{3s}}{s} = e^{3s}\left[\frac{1}{s(s+3)}\right].$$

Using partial fractions,

$$\frac{1}{s(s+3)} = \frac{1}{3}\left(\frac{1}{s} - \frac{1}{s+3}\right).$$

Therefore,

$$Y(s) = e^{3s}\left[\frac{1}{3}\left(\frac{1}{s} - \frac{1}{s+3}\right)\right].$$

Let

$$F(s) = \frac{1}{3}\left(\frac{1}{s} - \frac{1}{s+3}\right).$$

Its inverse Laplace transform is

$$f(t) = \frac{1}{3}\left(1 - e^{-3t}\right)u(t).$$

Since

$$Y(s) = e^{3s}F(s),$$

using the time-advance property,

$$y(t) = f(t+3).$$

Hence,

$$y(t) = \frac{1}{3}\left[1 - e^{-3(t+3)}\right]u(t+3).$$

As

$$t \to \infty,$$

$$e^{-3(t+3)} \to 0, \qquad u(t+3) \to 1.$$

Therefore,

$$y(\infty) = \frac{1}{3} = 0.33.$$

$$\boxed{0.33}$$

Key Points

- Convolution in time corresponds to multiplication in the Laplace domain.
- A factor e^{sT} in the Laplace domain represents a time advance by T.
- Evaluate the limit only after obtaining the complete time-domain expression.

Mistakes to be avoided

- Do not confuse the Laplace transform of $u(t+3)$ with that of $u(t-3)$.
- Do not forget to apply partial fraction decomposition before taking the inverse Laplace transform.
- Remember that $e^{-3(t+3)} \to 0$ as $t \to \infty$.

Q5.17.1

NAT | 2017 | 2M | Ans: -2.4 to -2.0 | ● ● ○ | CLICK FOR VIDEO SOLUTION

Theory: Express the Laplace transform in terms of standard transform pairs before taking the inverse Laplace transform.
Given,

$$Y(s) = \frac{s+2}{s+6}.$$

Rewrite it as

$$Y(s) = 1 - \frac{4}{s+6}.$$

Using standard Laplace transform pairs,

$$1 \xleftrightarrow{\mathcal{L}^{-1}} \delta(t),$$

and

$$\frac{1}{s+a} \xleftrightarrow{\mathcal{L}^{-1}} e^{-at}u(t),$$

we obtain

$$y(t) = \delta(t) - 4e^{-6t}u(t).$$

Since $t = 0.1 > 0$,

$$\delta(0.1) = 0,$$

thus,

$$y(0.1) = -4e^{-0.6}.$$

Evaluating,

$$y(0.1) = -4(0.5488) = -2.195.$$

Hence,

$$\boxed{-2.195}$$

Key Points

- Rewrite improper rational functions into standard Laplace transform forms.
- The inverse Laplace transform of 1 is $\delta(t)$.
- For $t > 0$, the impulse term contributes zero.

Mistakes to be avoided

- Do not forget to separate the constant term before inversion.
- Remember that $\delta(t)$ contributes only at $t = 0$.

- Evaluate the exponential term only after substituting the given time.

Q5.17.2 MCQ 2017 1M Ans: B ● ○ ○ CLICK FOR VIDEO SOLUTION

Theory: The transfer function is obtained by taking the Laplace transform of the differential equation under zero initial conditions. The inverse system has the reciprocal transfer function.
Taking the Laplace transform of

$$\frac{dy(t)}{dt} + 2y(t) = \frac{dx(t)}{dt} + x(t),$$

with zero initial conditions,

$$sY(s) + 2Y(s) = sX(s) + X(s).$$

Therefore,

$$(s+2)Y(s) = (s+1)X(s).$$

Hence, the transfer function of the system is

$$H(s) = \frac{Y(s)}{X(s)} = \frac{s+1}{s+2}.$$

The transfer function of the inverse system is

$$H^{-1}(s) = \frac{X(s)}{Y(s)} = \frac{s+2}{s+1}.$$

$$\boxed{\text{(B) } \frac{s+2}{s+1}}$$

Key Points

- Assume zero initial conditions while obtaining the transfer function.
- The inverse system has the reciprocal transfer function.
- Always derive $H(s)$ first, then invert it.

Mistakes to be avoided

- Do not interchange $X(s)$ and $Y(s)$ while forming the transfer function.
- Remember to include the derivative terms as multiplication by s.
- The inverse transfer function is the reciprocal of $H(s)$, not its negative.

Q5.16.1 MCQ 2016 1M Ans: 0 ● ○ ○ CLICK FOR VIDEO SOLUTION

Theory: The final value theorem gives the steady-state value of a signal provided all poles of $sX(s)$ lie in the left half-plane.
Using the final value theorem,

$$\lim_{t\to\infty} x(t) = \lim_{s\to 0} sX(s).$$

Substituting the given transform,

$$\lim_{t\to\infty} x(t) = \lim_{s\to 0} \frac{s(s+2)}{(s+1)(s+3)^2}.$$

Evaluating at $s = 0$,

$$\lim_{t\to\infty} x(t) = \frac{0\cdot 2}{1\cdot 9} = 0.$$

Hence,

$$\boxed{0}$$

Key Points

- Use the final value theorem only after verifying its applicability.
- Compute $\lim_{s\to 0} sX(s)$.
- All poles of $sX(s)$ must lie in the left half-plane.

Mistakes to be avoided

- Do not apply the theorem if right-half-plane or imaginary-axis poles are present.
- Multiply by s before taking the limit.
- Substitute $s = 0$ only after forming $sX(s)$.

Q5.13.1 MCQ 2013 2M Ans: D CLICK FOR VIDEO SOLUTION

Theory: A triangular waveform can be expressed as a combination of delayed ramp functions.
The given triangular pulse can be written as

$$x(t) = r(t) - 2r(t-1) + r(t-2),$$

where

$$r(t) = tu(t)$$

is the unit ramp.
Using the Laplace transform of the delayed ramp,

$$\mathcal{L}\{r(t-a)\} = \frac{e^{-as}}{s^2},$$

we obtain

$$X(s) = \frac{1}{s^2} - \frac{2e^{-s}}{s^2} + \frac{e^{-2s}}{s^2}.$$

Hence,

$$X(s) = \frac{1}{s^2}\left(1 - 2e^{-s} + e^{-2s}\right).$$

$$\boxed{\text{(D) } \frac{1}{s^2}\left(1 - 2e^{-s} + e^{-2s}\right)}$$

Key Points

- Represent triangular pulses using delayed ramp functions.
- $\mathcal{L}\{r(t-a)\} = \dfrac{e^{-as}}{s^2}$.
- Apply linearity of the Laplace transform after decomposition.

Mistakes to be avoided

- Do not represent the triangular waveform using step functions alone.
- Remember that delayed ramps introduce the exponential factor e^{-as}.
- Carefully preserve the coefficient -2 of the middle ramp.

Q5.12.1 MCQ | 2012 | 1M | Ans: D | CLICK FOR VIDEO SOLUTION

Theory: Multiplication by t in the time domain corresponds to differentiating the Laplace transform with respect to s.
Given,

$$F(s) = \frac{1}{s^2 + s + 1}.$$

Using the property,

$$\mathcal{L}\{tf(t)\} = -\frac{dF(s)}{ds}.$$

Differentiate $F(s)$,

$$\frac{dF(s)}{ds} = -\frac{2s + 1}{(s^2 + s + 1)^2}.$$

Therefore,

$$\mathcal{L}\{tf(t)\} = -\left(-\frac{2s+1}{(s^2+s+1)^2}\right) = \frac{2s+1}{(s^2+s+1)^2}.$$

Hence,

$$\boxed{\text{(D)}\ \frac{2s+1}{(s^2+s+1)^2}}$$

Key Points

- $\mathcal{L}\{tf(t)\} = -\frac{dF(s)}{ds}$.
- Differentiate the entire transform before applying the negative sign.
- Use the chain rule carefully while differentiating rational functions.

Mistakes to be avoided

- Do not forget the negative sign in the property.
- Differentiate the denominator correctly using the chain rule.
- Avoid sign errors while simplifying the final expression.

Q5.10.1 MCQ | 2010 | 1M | Ans: C | CLICK FOR VIDEO SOLUTION

Theory: A delayed unit-step function introduces an exponential factor in the Laplace domain.
Given,

$$f(t) = u(t - \tau).$$

Using the time-delay property,

$$\mathcal{L}\{f(t-\tau)u(t-\tau)\} = e^{-s\tau}F(s).$$

Since

$$\mathcal{L}\{u(t)\} = \frac{1}{s},$$

we obtain

$$\mathcal{L}\{u(t-\tau)\} = e^{-s\tau} \cdot \frac{1}{s} = \frac{e^{-s\tau}}{s}.$$

Hence,

$$\boxed{\text{(C)}\ \frac{e^{-s\tau}}{s}}$$

Key Points

- A time delay of τ introduces the factor $e^{-s\tau}$.
- $\mathcal{L}\{u(t)\} = \frac{1}{s}$.
- Apply the time-shifting property only after expressing the signal in delayed form.

Mistakes to be avoided

- Do not confuse $u(t-\tau)$ with $u(t)e^{-\tau t}$.
- Do not omit the exponential delay factor.
- Remember that the denominator remains s.

Q5.07.1 MCQ | 2007 | 2M | Ans: A | ● ○ ○ | CLICK FOR VIDEO SOLUTION

Theory: Time delay introduces a phase shift, while differentiation multiplies the spectrum by $j\omega$.
Given,

$$x(t) \overset{\mathcal{F}}{\longleftrightarrow} X(\omega).$$

Using the time-delay property,

$$x(t-t_d) \overset{\mathcal{F}}{\longleftrightarrow} e^{-j\omega t_d} X(\omega).$$

Applying the differentiation property,

$$\frac{d}{dt}x(t-t_d) \overset{\mathcal{F}}{\longleftrightarrow} j\omega\, e^{-j\omega t_d} X(\omega).$$

Taking magnitude,

$$|Y(\omega)| = |j\omega|\,|e^{-j\omega t_d}|\,|X(\omega)|.$$

Since

$$|j| = 1, \qquad |e^{-j\omega t_d}| = 1,$$

we obtain

$$|Y(\omega)| = |\omega|\,|X(\omega)|.$$

Hence,

$$\boxed{\text{(A) } |\omega|\,|X(\omega)|}$$

Key Points

- Differentiation multiplies the spectrum by $j\omega$.
- Time delay contributes only a phase factor of unit magnitude.
- The magnitude of $e^{-j\omega t_d}$ is always 1.

Mistakes to be avoided

- Do not include the delay term while computing magnitude.
- Use $|\omega|$, not ω, for the magnitude expression.
- Remember that $|j| = 1$.

Q5.05.1 MCQ 2005 1M Ans: A ● ○ ○ CLICK FOR VIDEO SOLUTION

Theory: An all-pass filter has a unity magnitude response for all frequencies while introducing only phase variation.
For the transfer function

$$H(s) = \frac{1 - s\tau}{1 + s\tau},$$

put

$$s = j\omega.$$

Then

$$H(j\omega) = \frac{1 - j\omega\tau}{1 + j\omega\tau}.$$

Hence,

$$|H(j\omega)| = \frac{\sqrt{1 + \omega^2\tau^2}}{\sqrt{1 + \omega^2\tau^2}} = 1,$$

for all values of ω.
Therefore, this transfer function is an all-pass filter.

$$\boxed{\text{(A) } \frac{1 - s\tau}{1 + s\tau}}$$

Key Points

- An all-pass filter has unity magnitude response.
- Only the phase of the signal is modified.
- Verify the magnitude by substituting $s = j\omega$.

Mistakes to be avoided

- Do not identify an all-pass filter from its appearance alone.
- Always check that $|H(j\omega)| = 1$ for all frequencies.
- Do not confuse an all-pass filter with a low-pass or high-pass filter.

DEBUG INFO

Chapter V

Laplace Transform

Total Questions : 17

MCQ : 15

MSQ : 0

NAT : 2

SOLUTIONS

VI

Z Transform

Topics

1. Introduction to Z-Transform
2. Z-Transform and ROC
3. Inverse Z-Transform
4. Important Properties of Z-Transform
5. Unilateral Z-Transform

Q6.24.1 NAT 2024 2M Ans: 0.75 ○ ○ ○

Given $y[n] - 0.5\,y[n-2] = a\,x[n-4]$. Taking the Z-transform,

$$Y(z) - 0.5z^{-2}Y(z) = aZ^{-4}X(z) \;\Rightarrow\; H(Z) = \frac{Y(Z)}{X(Z)} = \frac{aZ^{-4}}{1-0.5Z^{-2}}.$$

Substituting $z = e^{j\omega}$ for the frequency response,

$$H(e^{j\omega}) = \frac{ae^{-j4\omega}}{1-0.5e^{-j2\omega}}.$$

At $\omega = \frac{\pi}{2}$, given $\left|H(e^{j\pi/2})\right| = 0.5$:

$$0.5 = \left|\frac{a\,e^{-j2\pi}}{1-0.5e^{-j\pi}}\right| = \frac{a}{1-0.5(-1)} = \frac{a}{1.5}.$$

$$a = 1.5 \times 0.5 = 0.75.$$

$$\boxed{0.75}$$

Q6.24.2 MCQ 2024 2M Ans: C ● ● ○

Given

$$x(n) = \begin{cases} (0.2)^n, & 0 \le n \le 7 \\ 0, & \text{otherwise} \end{cases}$$

$$X(z) = \sum_{n=0}^{7}(0.2)^n z^{-n} = 1 + (0.2)z^{-1} + (0.2)^2 z^{-2} + \cdots + (0.2)^7 z^{-7}.$$

This is a finite sum of negative powers of z. It converges everywhere *except* $z = 0$, where the negative-power terms are undefined; there is no restriction at $z = \infty$ since the expression reduces to the finite constant term 1 there.
Hence, the ROC is all values of z except $z = 0$, and the correct option is **(C)**.

$$\boxed{\text{C) all values of } z \text{ except } z = 0}$$

Q6.23.1 MCQ 2023 2M Ans: B ● ● ○

Given $F(z) = \frac{1}{1-z}$, expanded around $z = 2$. Let $t = z - 2$, so $z = 2 + t$.

$$F(t) = \frac{1}{1-(2+t)} = \frac{1}{-1-t} = \frac{-1}{1+t} = -\left(1 - t + t^2 - t^3 + \cdots\right).$$

$$F(z) = -1 + (z-2) - (z-2)^2 + (z-2)^3 - \cdots = \sum_{k=0}^{\infty}(-1)^{k+1}(z-2)^k.$$

Comparing with $F(z) = \sum_{k=0}^{\infty} a_k (z-2)^k$, we get $a_k = (-1)^{k+1}$.

$$\boxed{\text{B) } (-1)^{k+1}}$$

Q6.18.1 MCQ | 2018 | 2M | Ans: D | ● ● ○

Given $Y(z) = 2 + 3z^{-1} + z^{-2}$ is the Z-transform of $y[n] = x[n] * h[n]$, and $p[n] = x[n] * h[n-2]$.

By the time-shifting property, convolving with a signal delayed by 2 samples multiplies the transform by z^{-2}:

$$P(z) = z^{-2}Y(z) = z^{-2}\left(2 + 3z^{-1} + z^{-2}\right) = 2z^{-2} + 3z^{-3} + z^{-4}.$$

$$\boxed{\text{D) } 2z^{-2} + 3z^{-3} + z^{-4}}$$

Q6.17.1 MCQ | 2017 | 1M | Ans: C | ● ○ ○

Given $x_1(n) = \{1, 1\}$ and $x_2(n) = \{1, 2\}$.

$$x_1(n) \xrightarrow{\text{Z.T.}} 1 + z^{-1}, \qquad x_2(n) \xrightarrow{\text{Z.T.}} 1 + 2z^{-1}.$$

By the convolution property,

$$x_1(n) \otimes x_2(n) = X_1(z)X_2(z) = (1 + z^{-1})(1 + 2z^{-1}) = 1 + 3z^{-1} + 2z^{-2}.$$

$$\boxed{\text{C) } 1 + 3z^{-1} + 2z^{-2}}$$

Q6.15.1 MCQ | 2015 | 1M | Ans: A | ● ○ ○

Given $x(n) = \alpha^{|n|}$, $0 < |\alpha| < 1$. This can be split as

$$x(n) = \underbrace{\alpha^n u(n)}_{x_1(n)} + \underbrace{\alpha^{-n} u(-n-1)}_{x_2(n)}.$$

Using the standard pairs,

$$\alpha^n u(n) \overset{\text{ZT}}{\longleftrightarrow} \frac{1}{1 - \alpha z^{-1}}, \quad \text{ROC} : |z| > |\alpha|,$$
$$-\left(\frac{1}{\alpha}\right)^n u(-n-1) \overset{\text{ZT}}{\longleftrightarrow} \frac{1}{1 - \frac{1}{\alpha}z^{-1}}, \quad \text{ROC} : |z| < \frac{1}{|\alpha|}.$$

The common ROC exists (since $|\alpha| < 1$) and equals

$$|\alpha| < |z| < \frac{1}{|\alpha|}.$$

$$\boxed{\text{A) } |\alpha| < |z| < \frac{1}{|\alpha|}}$$

.

Q6.14.1 MCQ | 2014 | 2M | Ans: C | ● ● ○

Given $\dfrac{b_0}{1 - z^{-1} + a_2 z^{-2}}$, a_2 real. Multiplying through by z^2, the poles satisfy

$$z^2 - z + a_2 = 0 \;\Rightarrow\; z = \frac{1 \pm \sqrt{1 - 4a_2}}{2}.$$

Checking each option for $|z| < 1$ (BIBO stability):

- $a_2 = 0.5$: $z = \frac{1 \pm j}{2}$, $|z| = \frac{1}{\sqrt{2}} \approx 0.707 < 1 \Rightarrow$ **stable**.
- $a_2 = 1.5$: $|z| = \frac{\sqrt{6}}{2} \approx 1.22 > 1 \Rightarrow$ unstable.
- $a_2 = -0.75$: $z = 1.5, -0.5$; $|1.5| > 1 \Rightarrow$ unstable.
- $a_2 = -1.5$: $z \approx 1.82, -0.82$; $|1.82| > 1 \Rightarrow$ unstable.

$$\boxed{\text{C) } 0.5}$$

Q6.14.2 MCQ 2014 2M Ans: C ●●○

Given $H(z) = \dfrac{1 - \frac{1}{3}z^{-1}}{1 - \frac{1}{4}z^{-1}}$. $H(z)$ has a pole at $z = \frac{1}{4}$ and a zero at $z = \frac{1}{3}$.

For $H(z)$ itself to be causal and stable, its ROC must lie outside its outermost pole:

$$|z| > \frac{1}{4}.$$

$$\boxed{\text{C) } |z| > \frac{1}{4}}$$

Q6.13.1 MCQ 2013 1M Ans: D ●○○

Given $\dfrac{1 - 2z^{-1}}{1 - 0.5z^{-1}}$.

Zero: $1 - 2z^{-1} = 0 \Rightarrow z = 2$, which lies *outside* the unit circle $\Rightarrow$ the system is **non-minimum phase**.
Pole: $1 - 0.5z^{-1} = 0 \Rightarrow z = 0.5$, which lies *inside* the unit circle $\Rightarrow$ the system is **stable**.

$$\boxed{\text{D) non-minimum phase and stable}}$$

Q6.12.1 MCQ 2012 2M Ans: C ●●○

Given $x[n] = \left(\frac{1}{3}\right)^{|n|} - \left(\frac{1}{2}\right)^{n} u[n]$.

The two-sided term splits as $\left(\frac{1}{3}\right)^{|n|} = \left(\frac{1}{3}\right)^{n} u[n] + \left(\frac{1}{3}\right)^{-n} u[-n-1]$, whose ROC is $\frac{1}{3} < |z| < 3$.

The term $-\left(\frac{1}{2}\right)^{n} u[n]$ is right-sided with ROC $|z| > \frac{1}{2}$.

The overall ROC is the intersection of the two:

$$\max\left(\frac{1}{3}, \frac{1}{2}\right) < |z| < 3 \implies \frac{1}{2} < |z| < 3.$$

$$\boxed{\text{C) } \frac{1}{2} < |z| < 3}$$

Q6.11.1 MCQ | 2011 | 2M | Ans: B | ● ● ○

Given $y(n) - \frac{1}{3}y(n-1) = x(n), x(n) = \left(\frac{1}{2}\right)^n u(n)$. Taking the Z-transform under initial-rest conditions,

$$Y(z) - \frac{1}{3}z^{-1}Y(z) = \frac{z}{z-\frac{1}{2}} \Rightarrow Y(z) = \frac{z}{\left(z-\frac{1}{2}\right)\left(1-\frac{1}{3}z^{-1}\right)} = \frac{z^2}{\left(z-\frac{1}{2}\right)\left(z-\frac{1}{3}\right)}.$$

Expanding $Y(z)/z$ in partial fractions,

$$\frac{Y(z)}{z} = \frac{3}{z-\frac{1}{2}} - \frac{2}{z-\frac{1}{3}} \Rightarrow Y(z) = \frac{3z}{z-\frac{1}{2}} - \frac{2z}{z-\frac{1}{3}}.$$

Taking the inverse Z-transform,

$$y(n) = -2\left(\frac{1}{3}\right)^n + 3\left(\frac{1}{2}\right)^n, \quad n \geq 0.$$

$$\boxed{\text{B) } -2\left(\frac{1}{3}\right)^n + 3\left(\frac{1}{2}\right)^n}$$

Q6.10.1 MCQ | 2010 | 2M | Ans: B | ● ● ○

For $H(z)$ to be stable, its poles must lie inside the unit circle. For the inverse system $1/H(z)$ to be stable, its poles — i.e., the *zeros* of $H(z)$ — must also lie inside the unit circle.
Hence both the poles *and* zeros of $H(z)$ must lie inside the unit circle, and the correct option is **(B)**.

$$\boxed{\text{B) poles and zeros must be inside the unit circle}}$$

Q6.08.1 MCQ | 2008 | 1M | Ans: A | ● ○ ○

Given $x(n) = 2^n u(n)$, a right-sided (causal) signal.

$$X(z) = \sum_{n=-\infty}^{\infty} x(n)z^{-n} = \sum_{n=0}^{\infty} 2^n z^{-n} = \sum_{n=0}^{\infty} \left(2z^{-1}\right)^n.$$

This converges only if $|2z^{-1}| < 1$, i.e. $|z| > 2$, giving

$$X(z) = \frac{1}{1-2z^{-1}} = \frac{z}{z-2}, \qquad \text{ROC} : |z| > 2.$$

$$\boxed{\text{A) } |z| > 2}$$

Q6.08.2 MCQ | 2008 | 2M | Ans: A | ● ● ○

Given $y(n) = x(n) - \frac{1}{3}y(n-1), x(n) = \delta(n)$, with $y(n) = 0$ for $n < 0$ (causal). Taking the Z-transform,

$$Y(z) = 1 - \frac{1}{3}z^{-1}Y(z) \Rightarrow Y(z)\left(1 + \frac{1}{3z}\right) = 1 \Rightarrow Y(z) = \frac{3z}{3z+1} = \frac{z}{z+\frac{1}{3}}.$$

Taking the inverse Z-transform,

$$y(n) = \left(-\frac{1}{3}\right)^n u(n).$$

$$\boxed{\text{A) } \left(-\frac{1}{3}\right)^n u[n]}$$

Q6.05.1 MCQ 2005 2M Ans: D

From the given signal flow graph (SFG), applying Mason's Gain Formula:

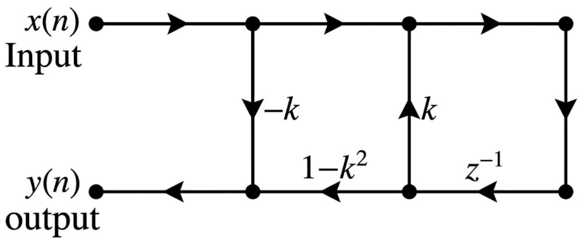

Fig. Solution Q6.05.1

$$H(z) = \frac{Y(z)}{X(z)} = \frac{\sum P_k \Delta_k}{\Delta}$$

1. Forward Path Gains (P_k):

- $P_1 = -k$
- $P_2 = (1-k^2)z^{-1}$

2. Individual Loop Gains (L_m):

- $L_1 = kz^{-1}$

3. Determinant (Δ) and Path Determinants (Δ_k):

$$\Delta = 1 - L_1 = 1 - kz^{-1}$$

- $\Delta_1 = 1 - kz^{-1}$ (Path P_1 does not touch Loop L_1)
- $\Delta_2 = 1$ (Path P_2 touches Loop L_1)

Substituting the values into the transfer function expression:

$$H(z) = \frac{P_1\Delta_1 + P_2\Delta_2}{\Delta} = \frac{-k\left(1-kz^{-1}\right) + (1-k^2)z^{-1}}{1-kz^{-1}}$$

$$= \frac{-k + k^2z^{-1} + z^{-1} - k^2z^{-1}}{1-kz^{-1}} = \frac{-k+z^{-1}}{1-kz^{-1}}$$

$$\boxed{\text{D) } \frac{-k+z^{-1}}{1-kz^{-1}}}$$

Q6.04.1 MCQ 2004 2M Ans: C

Given:

$$X(z) = \frac{1/2}{1-az^{-1}} + \frac{1/3}{1-bz^{-1}} \qquad \ldots(i)$$

where $|a| < 1$ and $|b| < 1$

ROC : $|a| < |z| < |b|$

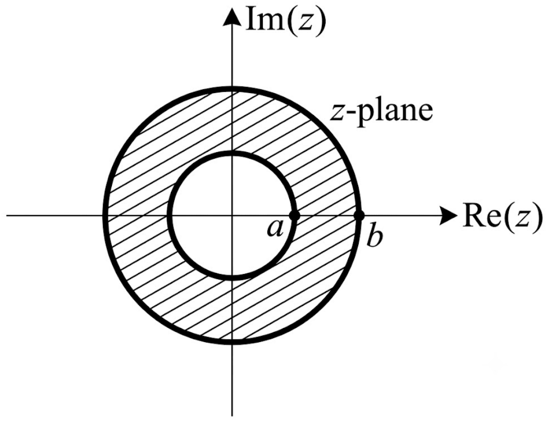

Fig. Solution Q6.04.1

According to given ROC,

$$X(z) = \underbrace{\frac{(1/2)z}{z-a}}_{\text{Right sided signal}} + \underbrace{\left(\frac{1}{3}\right)\frac{z}{z-b}}_{\text{Left sided signal}}$$

$$x(n) = \frac{1}{2}(a)^n u(n) - \frac{1}{3}(b)^n u(-n-1)$$

$$x(0) = \frac{1}{2}$$

$$\boxed{\text{(C) } \frac{1}{2}}$$

Key Points

- An ROC of the form $|a| < |z| < |b|$ represents a two-sided sequence consisting of a right-sided part for $|z| > |a|$ and a left-sided part for $|z| < |b|$.
- The inverse z-transform yields $x(0) = \frac{1}{2}$ since $u(0) = 1$ and $u(-1) = 0$.

Q6.04.2 MCQ 2004 1M Ans: A

From the given signal flow graph (SFG), applying Mason's Gain Formula:

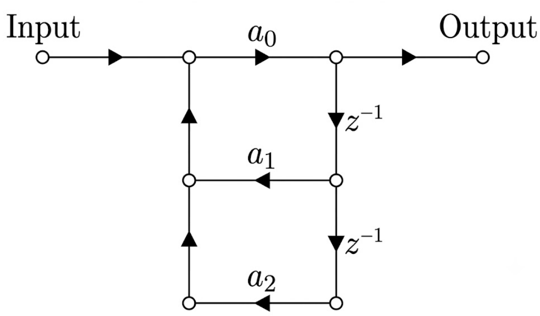

Fig. Solution Q6.04.2

$$H(z) = \frac{\sum P_k \Delta_k}{\Delta}$$

1. Forward Path Gain (P_k):

- Single forward path: $P_1 = 1 \cdot a_0 \cdot 1 = a_0$

2. Individual Loop Gains (L_m):

- Loop 1: $L_1 = z^{-1} \cdot a_1 = a_1 z^{-1}$
- Loop 2: $L_2 = z^{-1} \cdot z^{-1} \cdot a_2 = a_2 z^{-2}$

3. System Determinant (Δ) and Path Determinants (Δ_k):

$$\Delta = 1 - (L_1 + L_2) = 1 - a_1 z^{-1} - a_2 z^{-2}$$

Since P_1 touches both loops L_1 and L_2, we have $\Delta_1 = 1$.
4. Transfer Function Evaluation:

$$H(z) = \frac{a_0}{1 - a_1 z^{-1} - a_2 z^{-2}}$$

Comparing this with the given system function:

$$H(z) = \frac{1}{1 - 0.7z^{-1} + 0.13z^{-2}}$$

Equating the corresponding coefficients:

$$a_0 = 1.0, \qquad a_1 = 0.7, \qquad a_2 = -0.13$$

$$\boxed{\text{A) } 1.0,\ 0.7 \text{ and } -0.13}$$

Q6.04.3 MCQ 2004 2M Ans: D ● ● ○

Given: Discrete time sequence $x(n)$ suffered distortion modeled by an LTI system with,

$$H(z) = (1 - az^{-1}); \quad |a| > 1$$

To compensate the distortion, the system transfer function of a stable system should be $\frac{1}{H(z)}$.

$$\frac{1}{H(z)} = \frac{1}{1 - az^{-1}}$$

Inverse transform is given by,
For right sided signal,

$$h_c(n) = a^n u(n); \quad \text{ROC} : |z| > |a| \quad \ldots \text{(i)}$$

For left sided signal,

$$h_c(n) = -a^n u(-n-1); \quad \text{ROC} : |z| < |a| \quad \ldots \text{(ii)}$$

For a stable system, the ROC must include the unit circle ($|z| = 1$). Since $|a| > 1$, the ROC $|z| < |a|$ includes the unit circle, so equation (ii) represents the stable impulse response.

$$\boxed{\text{(D) } -a^n u(-n-1)}$$

Key Points

- An LTI system is BIBO stable if and only if the Region of Convergence (ROC) of its system function $H(z)$ contains the unit circle ($|z| = 1$).
- When a pole lies outside the unit circle ($|a| > 1$), a stable system requires a left-sided impulse response with ROC $|z| < |a|$.

Q6.03.1 MCQ 2003 1M Ans: A ● ○ ○

Given $X(z) = e^{1/z}$. Expanding as a power series in z^{-1},

$$X(z) = e^{1/z} = 1 + \frac{z^{-1}}{1!} + \frac{z^{-2}}{2!} + \frac{z^{-3}}{3!} + \cdots = \sum_{n=0}^{\infty} \frac{1}{n!} z^{-n}. \quad \text{(i)}$$

By definition,

$$X(z) = \sum_{n=-\infty}^{\infty} x(n)\, z^{-n}. \quad \text{(ii)}$$

Comparing (i) and (ii):

$$x(n) = \frac{1}{n!} \text{ for } n \geq 0 \quad \Rightarrow \quad x(n) = \frac{u(n)}{n!}.$$

$$\boxed{\text{A) } \frac{1}{n!} u[n]}$$

Q6.03.2 MCQ 2003 2M Ans: A ● ● ○

Given input $x = [a, b, c, d]$ and output $y = [x, x, x, x, \ldots \langle \text{repeated } N \text{ times} \rangle]$, i.e.

$$x(n) = a\delta(n) + b\delta(n-1) + c\delta(n-2) + d\delta(n-3),$$

$$y(n) = a\delta(n) + b\delta(n-1) + c\delta(n-2) + d\delta(n-3) + a\delta(n-4) + b\delta(n-5) + \cdots$$

The output repeats after every 4th sample, so the fundamental period is $N = 4$, and

$$y(n) = \sum_{i=0}^{N-1} x(n-4i) \quad \Longrightarrow \quad Y(z) = \sum_{i=0}^{N-1} X(z)\, z^{-4i}.$$

$$H(z) = \frac{Y(z)}{X(z)} = \sum_{i=0}^{N-1} z^{-4i} = 1 + z^{-4} + z^{-8} + \cdots + z^{-4(N-1)},$$

$$h(n) = \sum_{i=0}^{N-1} \delta(n-4i).$$

$$\boxed{\text{A) } \sum_{i=0}^{N-1} \delta[n-4i]}$$

Q6.01.1 MCQ | 2001 | 1M | Ans: A | ● ○ ○

Given $x(n) = \delta(n) \xrightarrow{h(n)} y(n)$. By the convolution property, $y(n) = x(n) \otimes h(n) \xrightarrow{\text{Z.T.}} Y(z) = X(z)H(z)$. With $x(n) = \delta(n) \leftrightarrow 1$ and $h(n) = a^n u(n) \leftrightarrow \dfrac{z}{z-a}$,

$$H(z) = \frac{Y(z)}{X(z)} = \frac{z}{z-a}.$$

For a unit-step input $x_1(n) = u(n) \leftrightarrow \dfrac{z}{z-1}$, ROC : $|z| > 1$.

$$Y_1(z) = \frac{z}{z-1} \cdot \frac{z}{z-a} = \frac{1}{1-a} \cdot \frac{z}{z-1} + \frac{a}{a-1} \cdot \frac{z}{z-a}.$$

Taking the inverse Z-transform,

$$y(n) = \frac{1}{1-a}u(n) + \frac{a}{a-1}a^n u(n) = \sum_{j=0}^{n} a^j.$$

$$\boxed{\text{A) } \sum_{j=0}^{n} a^j}$$

DEBUG INFO

Chapter VI

Z Transform

Total Questions : 21

MCQ : 20

MSQ : 0

NAT : 1

SOLUTIONS

VII

DTFT, DTFS, DFT & Miscellaneous

Topics

1. DTFT
2. Discrete-Time Fourier Series (DTFS)
3. Circular Convolution
4. DFT
5. Group Delay, Phase Delay and Basics of FFT

Q7.25.1 NAT 2025 2M Ans: 5

The given sequence is

$$x[0] = 1, \quad x[1] = 2, \quad x[2] = 2, \quad x[3] = 4.$$

The DTFT is sampled at

$$\omega_k = \frac{2\pi k}{3}, \qquad k = 0, 1, 2,$$

to obtain

$$Y[k] = X\left(e^{j\omega_k}\right).$$

Using the inverse 3-point DFT,

$$y[n] = \frac{1}{3}\sum_{k=0}^{2} Y[k]e^{j\frac{2\pi kn}{3}}.$$

For $n = 0$,

$$y[0] = \frac{1}{3}\sum_{k=0}^{2} Y[k].$$

Using the aliasing property of DTFT sampling,

$$y[n] = \sum_{m=-\infty}^{\infty} x[n + 3m].$$

Hence,

$$y[0] = x[0] + x[3] = 1 + 4 = 5.$$

$$\boxed{5}$$

Q7.24.1 NAT 2024 1M Ans: 32 CLICK FOR VIDEO SOLUTION

Theory: In a radix-2 DIT FFT, the number of complex multiplications required is $\frac{N}{2}\log_2 N$.
For a 16-point FFT,

$$N = 16, \qquad \log_2 16 = 4.$$

Hence, the number of complex multiplications is

$$\frac{N}{2}\log_2 N = \frac{16}{2} \times 4 = 8 \times 4 = 32.$$

Hence,

$$\boxed{32}$$

Key Points

- Radix-2 DIT FFT requires $\frac{N}{2}\log_2 N$ complex multiplications.
- The number of stages in radix-2 FFT is $\log_2 N$.
- Each stage contains $\frac{N}{2}$ butterfly operations.

Mistakes to be avoided

- Do not use the formula $N\log_2 N$, which corresponds to the number of complex additions.
- Ensure that N is a power of 2 before applying the radix-2 FFT formula.

- Do not confuse butterfly operations with complex multiplications.

Q7.24.2 NAT 2024 2M Ans: 2 CLICK FOR VIDEO SOLUTION

Theory: Circular convolution in the time domain corresponds to pointwise multiplication in the DFT domain.
Using the circular convolution property,

$$Z[k] = X[k]Y[k].$$

Hence,

$$\begin{aligned} Z[0] &= 1 \cdot 1 = 1, \\ Z[1] &= (-j)(3j) = 3, \\ Z[2] &= 1 \cdot 1 = 1, \\ Z[3] &= (j)(-3j) = 3. \end{aligned}$$

Thus,

$$Z[k] = [1, 3, 1, 3].$$

Applying the IDFT,

$$z[0] = \frac{1}{4}\sum_{k=0}^{3} Z[k] = \frac{1}{4}(1+3+1+3) = 2.$$

Hence,

$$\boxed{z[0] = 2}$$

Key Points

- Circular convolution in time domain becomes multiplication in the DFT domain.
- The IDFT reconstructs the time-domain sequence from its DFT.
- The first sample is the average of all DFT coefficients.

Mistakes to be avoided

- Do not perform linear convolution instead of circular convolution.
- Remember to multiply the DFT coefficients element-wise before taking the IDFT.
- Do not forget the $\frac{1}{N}$ scaling factor in the IDFT.

Q7.20.1 NAT 2020 2M Ans: 0 CLICK FOR VIDEO SOLUTION

Theory: The IDFT reconstructs the time-domain sequence from its frequency-domain samples.
Given,

$$X(0) = X(1) = X(2) = 1, \qquad N = 3.$$

The IDFT is

$$x(n) = \frac{1}{N}\sum_{k=0}^{N-1} X(k)e^{j\frac{2\pi}{N}kn}.$$

For $n = 2$,

$$x(2) = \frac{1}{3}\sum_{k=0}^{2} e^{j\frac{4\pi}{3}k} = \frac{1}{3}\left(1 + e^{j\frac{4\pi}{3}} + e^{j\frac{8\pi}{3}}\right).$$

Since

$$e^{j\frac{8\pi}{3}} = e^{j\frac{2\pi}{3}},$$

we get

$$x(2) = \frac{1}{3}\left[1 + \left(-\frac{1}{2} - j\frac{\sqrt{3}}{2}\right) + \left(-\frac{1}{2} + j\frac{\sqrt{3}}{2}\right)\right].$$

Hence,

$$x(2) = \frac{1}{3}(1 - 1) = 0.$$

$$\boxed{\text{x(2)=0}}$$

Key Points

- The IDFT is given by $x(n) = \dfrac{1}{N}\sum_{k=0}^{N-1} X(k)e^{j\frac{2\pi}{N}kn}$.
- The cube roots of unity satisfy $1 + e^{j2\pi/3} + e^{j4\pi/3} = 0$.
- Complex exponential periodicity, $e^{j(\theta+2\pi m)} = e^{j\theta}$, simplifies the computation.

Mistakes to be avoided

- Do not omit the normalization factor $\dfrac{1}{N}$ in the IDFT.
- Do not use the DFT exponent $(-j)$ instead of the IDFT exponent $(+j)$.
- Remember that $e^{j8\pi/3} = e^{j2\pi/3}$ due to periodicity.

Q7.19.1 NAT | 2019 | 2M | Ans: 8 | ● ○ ○ | CLICK FOR VIDEO SOLUTION

Theory: The normalized digital frequency and analog frequency are related by

$$\omega = 2\pi\frac{f}{f_s}.$$

Given,

$$\omega_p = 0.3\pi,$$

and

$$f_p = 1.2\text{ kHz}.$$

Using

$$\omega_p = 2\pi\frac{f_p}{f_s},$$

we obtain

$$0.3\pi = 2\pi\frac{1.2}{f_s}.$$

Hence,

$$f_s = \frac{2\pi(1.2)}{0.3\pi} = 8\text{ kHz}.$$

Therefore,

$$\boxed{8\text{ kHz}}$$

Key Points

- Use the relation $\omega = 2\pi\dfrac{f}{f_s}$.

- Normalized frequency is expressed in rad/sample.
- Use the passband edge frequency for determining the sampling frequency.

Mistakes to be avoided

- Do not confuse normalized frequency with analog frequency.
- Ensure that both frequencies are expressed in consistent units.
- Cancel the common factor of π before solving for f_s.

Q7.18.1 MCQ 2018 2M Ans: C CLICK FOR VIDEO SOLUTION

Theory: The 4-point DFT can be evaluated using the DFT matrix.
For $N = 4$,

$$W = e^{-j\frac{2\pi}{4}} = -j.$$

The DFT is

$$X[k] = \begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & -j & -1 & j \\ 1 & -1 & 1 & -1 \\ 1 & j & -1 & -j \end{bmatrix} \begin{bmatrix} 1 \\ -1 \\ 1 \\ -1 \end{bmatrix}.$$

Evaluating each coefficient,

$$X(0) = 1 - 1 + 1 - 1 = 0,$$
$$X(1) = 1 + j - 1 - j = 0,$$
$$X(2) = 1 + 1 + 1 + 1 = 4,$$
$$X(3) = 1 - j - 1 + j = 0.$$

Thus,

$$X(2) = 4,$$

and all other coefficients are zero.
Therefore,

$$\boxed{\text{(C) } 2}$$

Key Points

- For a 4-point DFT, $W = e^{-j\pi/2} = -j$.
- The given sequence alternates in sign, concentrating its spectrum at $k = 2$.
- Evaluate each DFT coefficient systematically using the DFT matrix or definition.

Mistakes to be avoided

- Do not confuse the value of k with the value of $X(k)$.
- Use the correct twiddle factor $W = -j$.
- Carefully handle the complex arithmetic while evaluating $X(1)$ and $X(3)$.

Q7.17.1 MCQ 2017 2M Ans: B CLICK FOR VIDEO SOLUTION

Theory: The DFT of a real sequence satisfies conjugate symmetry.
For a real sequence,

$$X(k) = X^*(N-k).$$

Since

$$N = 5,$$

we obtain

$$X(2) = X^*(3) = 2 - j2,$$

and

$$X(4) = X^*(1) = 1 + j.$$

Using the IDFT,

$$x(0) = \frac{1}{5}\sum_{k=0}^{4} X(k).$$

Substituting the values,

$$\begin{aligned} x(0) &= \frac{1}{5}[0 + (1-j) + (2-j2) + (2+j2) + (1+j)] \\ &= \frac{1}{5}(10) \\ &= 2. \end{aligned}$$

$$\boxed{\text{(B) } 2}$$

Key Points

- For real sequences, use conjugate symmetry to determine missing DFT coefficients.
- At $n = 0$, the IDFT reduces to the average of all DFT coefficients.
- Imaginary parts cancel due to conjugate symmetry.

Mistakes to be avoided

- Do not assume the missing coefficients are zero.
- Use $X(k) = X^*(N-k)$ correctly for odd N.
- Divide the total sum by $N = 5$.

Q7.15.1 MCQ 2015 2M Ans: D CLICK FOR VIDEO SOLUTION

Theory: The sequence

$$\frac{\sin(\omega_c n)}{\pi n}$$

is the impulse response of an ideal low-pass filter with cutoff frequency ω_c.
Given,

$$x[n] = \frac{\sin\left(\frac{\pi n}{6}\right)}{\pi n},$$

and

$$h[n] = \frac{\sin(\omega_c n)}{\pi n}, \qquad \omega_c > \frac{\pi}{6}.$$

The spectrum of $x[n]$ occupies

$$|\omega| \le \frac{\pi}{6},$$

while the filter passband is

$$|\omega| \le \omega_c, \qquad \omega_c > \frac{\pi}{6}.$$

Hence, the entire spectrum of $x[n]$ lies inside the passband of the filter.
Therefore,

$$Y(e^{j\omega}) = X(e^{j\omega}),$$

which implies

$$y[n] = x[n].$$

Thus,

$$y[n] = \frac{\sin\left(\frac{\pi n}{6}\right)}{\pi n}.$$

$$\boxed{\text{(D) } \frac{\sin\left(\frac{\pi n}{6}\right)}{\pi n}}$$

Key Points

- $\dfrac{\sin(\omega_c n)}{\pi n}$ is the impulse response of an ideal low-pass filter.
- If the signal bandwidth is smaller than the filter cutoff frequency, the signal passes unchanged.
- Multiplication in the frequency domain determines the output spectrum.

Mistakes to be avoided

- Do not perform convolution directly in the time domain for this problem.
- Verify that the signal bandwidth lies entirely within the filter passband.
- Do not confuse the cutoff frequency ω_c with the signal bandwidth $\pi/6$.

Q7.15.2 MCQ 2015 2M Ans: A CLICK FOR VIDEO SOLUTION

Theory: In a cascade connection, the overall magnitude response is the product of the individual magnitude responses.
For one filter,

$$|H(\omega)| = 1 - \delta$$

at the pass-band edge.
For M identical cascaded filters,

$$|H_{\text{eq}}(\omega)| = (1-\delta)^M.$$

Hence, the effective pass-band ripple is

$$\delta_{\text{eq}} = 1 - (1-\delta)^M.$$

Therefore,

$$\boxed{\text{(A) } 1 - (1-\delta)^M}$$

Key Points

- Magnitude responses multiply when filters are cascaded.
- Pass-band gain after cascading is $(1-\delta)^M$.

- Ripple is measured relative to unity gain.

Mistakes to be avoided

- Do not multiply ripple values directly.
- Ripple is computed from the overall gain, not by adding individual ripples.
- Remember that cascading multiplies transfer functions, not their deviations.

Q7.15.3 MCQ 2015 1M Ans: D CLICK FOR VIDEO SOLUTION

Theory: For a linear-phase FIR filter with real coefficients, if z_0 is a zero, then the following are also zeros:

$$z_0^*, \quad \frac{1}{z_0}, \quad \frac{1}{z_0^*}.$$

Given,

$$z_0 = 3 + 4j.$$

Its complex conjugate is

$$z_0^* = 3 - 4j.$$

Its reciprocal is

$$\frac{1}{z_0} = \frac{1}{3+4j} = \frac{3-4j}{25} = \frac{3}{25} - \frac{4}{25}j.$$

The reciprocal of the conjugate is

$$\frac{1}{z_0^*} = \frac{1}{3-4j} = \frac{3+4j}{25} = \frac{3}{25} + \frac{4}{25}j.$$

Hence,

$$\frac{1}{3} - \frac{1}{4}j$$

does not belong to the required set of zeros.

$$\boxed{\text{(D) } \frac{1}{3} - \frac{1}{4}j}$$

Key Points

- A linear-phase FIR filter with real coefficients has zeros in conjugate and reciprocal-conjugate pairs.
- The four associated zeros are $z_0, \; z_0^*, \; \frac{1}{z_0}, \; \frac{1}{z_0^*}$.
- Always rationalize the denominator while computing reciprocals.

Mistakes to be avoided

- Do not confuse reciprocal with complex conjugate.
- Do not forget the reciprocal-conjugate property of linear-phase FIR filters.
- Compute $\dfrac{1}{3+4j}$ using complex conjugation of the denominator.

Q7.14.1 MCQ 2014 2M Ans: A CLICK FOR VIDEO SOLUTION

Theory: For a real-valued sequence,

$$X(k) = X^*(N-k).$$

Also,

$$x(n) = \frac{1}{N}\sum_{k=0}^{N-1} X(k)e^{j\frac{2\pi kn}{N}}.$$

Given,

$$N = 6,$$

$$X(0) = 9, \qquad X(2) = 2 + j2, \qquad X(3) = 3, \qquad X(5) = 1 - j.$$

Using conjugate symmetry,

$$X(1) = X^*(5) = 1 + j,$$

$$X(4) = X^*(2) = 2 - j2.$$

For $n = 0$,

$$x(0) = \frac{1}{6}\sum_{k=0}^{5} X(k).$$

Therefore,

$$\begin{aligned} x(0) &= \frac{1}{6}\left[9 + (1+j) + (2+j2) + 3 + (2-j2) + (1-j)\right] \\ &= \frac{1}{6}(18) \\ &= 3. \end{aligned}$$

$$\boxed{\text{(A) } 3}$$

Key Points

- For real sequences, $X(k) = X^*(N-k)$.
- At $n = 0$, all IDFT exponential terms become unity.
- Sum all DFT coefficients before dividing by N.

Mistakes to be avoided

- Do not forget to determine the missing DFT coefficients using conjugate symmetry.
- Remember that $X(0)$ and $X(N/2)$ are purely real for real sequences.
- Divide the final sum by N.

Q7.14.2 MCQ 2014 1M Ans: B CLICK FOR VIDEO SOLUTION

Theory: The frequency response is obtained by taking the DTFT of the impulse response.
Given,

$$h[n] = 0.5\left[\delta[n] + \delta[n-1]\right].$$

Hence,

$$H\left(e^{j\Omega}\right) = 0.5\left(1 + e^{-j\Omega}\right).$$

Its magnitude is

$$\left|H\left(e^{j\Omega}\right)\right| = \frac{1}{2}\sqrt{(1+\cos\Omega)^2 + \sin^2\Omega}.$$

Using

$$(1+\cos\Omega)^2+\sin^2\Omega = 2+2\cos\Omega = 4\cos^2\left(\frac{\Omega}{2}\right),$$

we obtain

$$\left|H\left(e^{j\Omega}\right)\right| = \cos\left(\frac{\Omega}{2}\right).$$

At the 3 dB cut-off frequency,

$$\left|H\left(e^{j\Omega_c}\right)\right| = \frac{1}{\sqrt{2}}.$$

Therefore,

$$\cos\left(\frac{\Omega_c}{2}\right) = \frac{1}{\sqrt{2}}$$

which gives

$$\frac{\Omega_c}{2} = \frac{\pi}{4} \quad \Rightarrow \quad \Omega_c = \frac{\pi}{2}.$$

Since

$$2\pi \text{ rad/sample} \longleftrightarrow 10 \text{ kHz},$$

the corresponding analog frequency is

$$f_c = \frac{\Omega_c}{2\pi} f_s = \frac{\pi/2}{2\pi} \times 10 = 2.5 \text{ kHz}.$$

Hence,

$$\boxed{\text{(B) } 2.50 \text{ kHz}}$$

Key Points

- Obtain the DTFT of the impulse response before finding the cut-off frequency.
- Use the identity $2+2\cos\Omega = 4\cos^2(\Omega/2)$.
- Convert normalized frequency to analog frequency using $f = \dfrac{\Omega}{2\pi} f_s$.

Mistakes to be avoided

- Do not equate the 3 dB point to $\Omega = \pi/4$; the correct value is $\Omega = \pi/2$.
- Do not forget to convert from rad/sample to Hz.
- Use the magnitude response, not the complex frequency response, for determining the cut-off frequency.

Q7.11.1 | MCQ | 2011 | 2M | Ans: A | ● ○ ○ | CLICK FOR VIDEO SOLUTION

Theory: Mason's Gain Formula is used to determine the transfer function of a signal flow graph.
The forward paths are

$$P_1 = -\frac{1}{s}, \qquad P_2 = -k.$$

There is one loop having loop gain

$$L = -\frac{k}{s}.$$

Hence,

$$\Delta = 1 - \sum L = 1 + \frac{k}{s}.$$

Since there are no non-touching loops,

$$\Delta_1 = \Delta_2 = 1.$$

Using Mason's Gain Formula,

$$H(s) = \frac{P_1\Delta_1 + P_2\Delta_2}{\Delta} = \frac{-\frac{1}{s} - k}{1 + \frac{k}{s}}.$$

Multiplying numerator and denominator by s,

$$H(s) = -\frac{1 + ks}{s + k}.$$

Therefore,

$$\boxed{\text{(A)} \ -\frac{1 + ks}{s + k}}$$

Key Points

- Apply Mason's Gain Formula systematically.
- Compute the determinant Δ using loop gains.
- Multiply numerator and denominator by s to simplify the final expression.

Mistakes to be avoided

- Do not miss any forward path while applying Mason's formula.
- Pay careful attention to the sign of each loop gain.
- Check whether non-touching loops exist before computing Δ.

Q7.11.2

MCQ | 2011 | 2M | Ans: D | ● ○ ○ | CLICK FOR VIDEO SOLUTION

Theory: An FIR filter has all its poles located at the origin (apart from possible pole-zero cancellations), whereas an IIR filter has poles at locations other than the origin.
From the pole-zero plot,

- There is a sixth-order pole at the origin.
- All remaining marked locations are zeros.

Hence, the transfer function can be written as

$$H(z) = \frac{(z-1)(z+1)(z-2)(z+2)\left(z-\frac{1}{2}\right)\left(z+\frac{1}{2}\right)}{z^6}.$$

Expanding the numerator,

$$H(z) = \frac{(z^2-4)\left(z^2-\frac{1}{4}\right)(z^2-1)}{z^6} = 1 - \frac{21}{4}z^{-2} + \frac{21}{4}z^{-4} - z^{-6}.$$

Taking the inverse z-transform,

$$h[n] = \delta[n] - \frac{21}{4}\delta[n-2] + \frac{21}{4}\delta[n-4] - \delta[n-6].$$

Since the impulse response has only a finite number of non-zero samples, the filter is an FIR filter.

$$\boxed{\text{(D) This is an FIR filter.}}$$

Key Points

- FIR filters have poles only at the origin.
- A finite-duration impulse response implies an FIR filter.
- Zeros can occur anywhere in the z-plane without affecting the FIR property.

Mistakes to be avoided

- Do not classify the filter as IIR merely because several zeros are present.
- Filter type (FIR/IIR) is determined by pole locations, not zero locations.
- Do not infer low-pass or high-pass behaviour solely from the pole-zero plot unless frequency response is analyzed.

Q7.10.1 MCQ 2010 2M Ans: A ● ○ ○ CLICK FOR VIDEO SOLUTION

Theory: For a real-valued sequence,

$$X[N-k] = X^*[k].$$

Since

$$N = 4,$$

we obtain

$$X[3] = X^*[1] = 1 - j.$$

Using the IDFT,

$$x[n] = \frac{1}{N}\sum_{k=0}^{N-1} X[k]e^{j\frac{2\pi kn}{N}}.$$

For $n = 0$,

$$x[0] = \frac{1}{4}\left(X[0] + X[1] + X[2] + X[3]\right).$$

Substituting the values,

$$x[0] = \frac{1}{4}[5 + (1+j) + 0 + (1-j)] = \frac{7}{4} = 1.75.$$

Since the DFT must satisfy conjugate symmetry for a real sequence,

$$X[3] = 1 - j.$$

Among the given options, only

$$(1 - j,\ 1.875)$$

matches the answer.

$$\boxed{\text{(A) } 1 - j,\ 1.875}$$

Key Points

- For real sequences, $X[N-k] = X^*[k]$.
- Use the IDFT formula to compute time-domain samples.
- For $n = 0$, all exponential terms are equal to unity.

Mistakes to be avoided

- Do not forget the conjugate symmetry property.
- Remember that $e^{j0} = 1$.

- Verify the arithmetic carefully when summing complex numbers.

Q7.10.2 MCQ 2010 1M Ans: B ● ○ ○ CLICK FOR VIDEO SOLUTION

Theory: In Direct Form-I realization, the feedforward and feedback sections are implemented separately, whereas Direct Form-II shares the delay elements between the two sections.
Given,

$$H(z) = \frac{P_0 + P_1 z^{-1} + P_3 z^{-3}}{1 + d_3 z^{-3}}.$$

Express the transfer function as

$$H(z) = H_1(z) H_2(z),$$

where

$$H_1(z) = P_0 + P_1 z^{-1} + P_3 z^{-3},$$

and

$$H_2(z) = \frac{1}{1 + d_3 z^{-3}}.$$

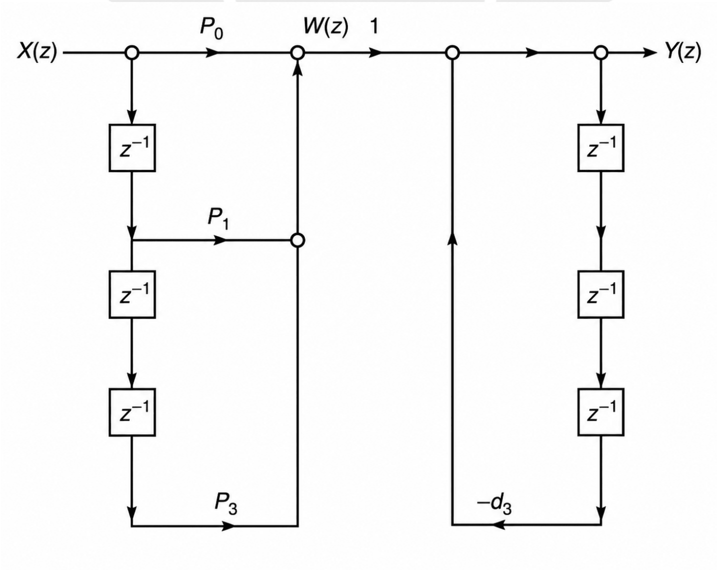

Fig. Solution Q7.10.2

For the feedforward section,

$$H_1(z),$$

the highest delay is

$$z^{-3},$$

hence it requires 3 delay elements.
Similarly, the feedback section,

$$H_2(z),$$

also requires

$$3$$

delay elements.
Therefore, Direct Form-I requires

$$3 + 3 = 6$$

delay elements.
In Direct Form-II realization, both sections share the same delay line. Hence, only the maximum order of the system is required.

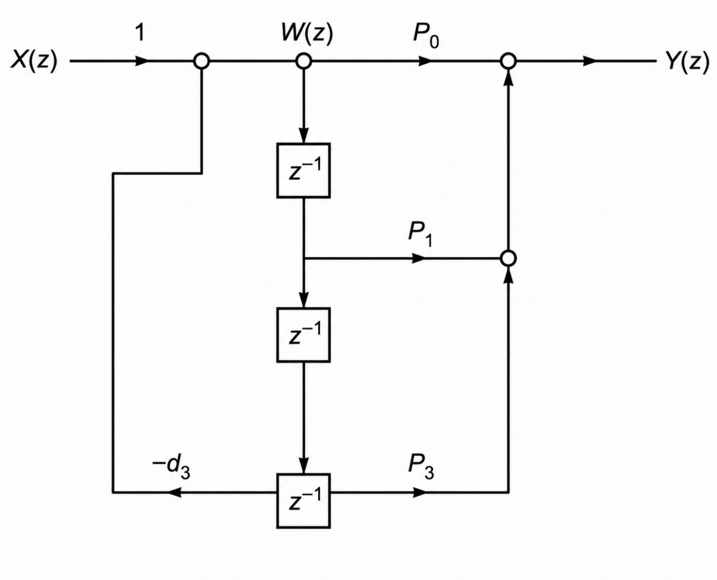

Fig. Solution Q7.10.2

Since the overall order is 3, so Direct Form-II requires 3 delay elements.
Therefore,

$$\boxed{\text{(B) 6 and 3}}$$

Key Points

- Direct Form-I uses separate delay lines for numerator and denominator sections.
- Direct Form-II shares the delay elements between the two sections.
- The number of delays in Direct Form-II equals the overall system order.

Mistakes to be avoided

- Do not count only the non-zero coefficients; count the highest delay order.
- Do not assume Direct Form-I and Direct Form-II require the same memory.
- Remember that Direct Form-II minimizes memory by sharing delay elements.

Q7.09.1 MCQ | 2009 | 2M | Ans: D | ● ○ ○ | CLICK FOR VIDEO SOLUTION

Theory: For an LTI system, a sinusoidal input undergoes only amplitude scaling and phase shifting. Given,

$$H(j\omega) = \frac{1}{\sqrt{1+\omega^2}} \angle \left(-\tan^{-1}\omega\right),$$

and

$$x(t) = \sin t.$$

The input frequency is

$$\omega = 1 \text{ rad/s}.$$

Hence,

$$|H(j1)| = \frac{1}{\sqrt{1+1^2}} = \frac{1}{\sqrt{2}},$$

and

$$\angle H(j1) = -\tan^{-1}(1) = -\frac{\pi}{4}.$$

Therefore,

$$y(t) = \frac{1}{\sqrt{2}}\sin\left(t - \frac{\pi}{4}\right).$$

$$\boxed{\text{(D) } \frac{1}{\sqrt{2}}\sin\left(t - \frac{\pi}{4}\right)}$$

Key Points

- A sinusoidal input remains sinusoidal after passing through an LTI system.
- Output amplitude is multiplied by $|H(j\omega)|$.
- Output phase is shifted by $\angle H(j\omega)$.

Mistakes to be avoided

- Do not change the input frequency.
- Evaluate the frequency response at the input frequency only.
- Apply both magnitude scaling and phase shift.

Q7.09.2 MCQ 2009 2M Ans: C ● ○ ○ CLICK FOR VIDEO SOLUTION

Theory: The frequency response is obtained by substituting $z = e^{j\omega}$.
Given,

$$H(z) = 1 - z^{-6}.$$

Therefore,

$$H(e^{j\omega}) = 1 - e^{-j6\omega}.$$

Its magnitude is

$$|H(e^{j\omega})| = \sqrt{(1-\cos 6\omega)^2 + \sin^2 6\omega}.$$

Using

$$(1-\cos 6\omega)^2 + \sin^2 6\omega = 2 - 2\cos 6\omega = 4\sin^2 3\omega,$$

we obtain

$$|H(e^{j\omega})| = 2|\sin 3\omega|.$$

For the magnitude response to be zero,

$$\sin 3\omega = 0,$$

which gives

$$3\omega = n\pi.$$

The digital and analog frequencies are related by

$$\omega = \frac{2\pi f}{f_s}.$$

Substituting $f_s = 9$ kHz,

$$3\left(\frac{2\pi f}{9000}\right) = n\pi,$$

which gives

$$f = \frac{n \times 9000}{6}.$$

For $n = 1$,

$$f = 1500 \text{ Hz} = 1.5 \text{ kHz}.$$

Hence,

$$\boxed{\text{(C) } 1.5 \text{ kHz}}$$

Key Points

- Evaluate the frequency response using $z = e^{j\omega}$.
- Use the identity $2 - 2\cos\theta = 4\sin^2(\theta/2)$.
- Relate digital and analog frequencies using $\omega = \dfrac{2\pi f}{f_s}$.

Mistakes to be avoided

- Do not forget to take the magnitude before equating it to zero.
- Use the sampling frequency while converting from ω to f.
- Remember that the smallest positive solution corresponds to $n = 1$.

Q7.09.3

MCQ | 2009 | 2M | Ans: A | ● ○ ○ | CLICK FOR VIDEO SOLUTION

Theory: A linear-phase FIR filter has a symmetric impulse response,

$$h[n] = h[M - 1 - n],$$

where M is the number of filter coefficients.
Let

$$G(z) = A_0 + A_1 z^{-1} + A_2 z^{-2}.$$

Then,

$$H(z) = (1 + 2z^{-1} + 3z^{-2})(A_0 + A_1 z^{-1} + A_2 z^{-2}).$$

Expanding,

$$H(z) = A_0 + (2A_0 + A_1)z^{-1} + (3A_0 + 2A_1 + A_2)z^{-2} + (3A_1 + 2A_2)z^{-3} + 3A_2 z^{-4}.$$

Hence,

$$\begin{aligned}
h[0] &= A_0,\\
h[1] &= 2A_0 + A_1,\\
h[2] &= 3A_0 + 2A_1 + A_2,\\
h[3] &= 3A_1 + 2A_2,\\
h[4] &= 3A_2.
\end{aligned}$$

Since the filter is fourth order,

$$M = 5.$$

Using symmetry,

$$h[0] = h[4] \quad \Rightarrow \quad A_0 = 3A_2,$$

and

$$h[1] = h[3]$$

gives

$$2A_0 + A_1 = 3A_1 + 2A_2.$$

Substituting

$$A_0 = 3A_2,$$

$$6A_2 + A_1 = 3A_1 + 2A_2,$$

$$2A_2 = A_1.$$

Choosing

$$A_2 = 1,$$

gives

$$A_1 = 2, \qquad A_0 = 3.$$

Therefore,

$$G(z) = 3 + 2z^{-1} + z^{-2}.$$

$$\boxed{\text{(A) } 3 + 2z^{-1} + z^{-2}}$$

Key Points

- A linear-phase FIR filter has a symmetric impulse response.
- Expand the polynomial first to obtain the impulse-response coefficients.
- Use symmetry equations to determine the unknown coefficients.

Mistakes to be avoided

- Do not equate polynomial coefficients directly without expansion.
- Remember that a fourth-order FIR filter has 5 coefficients.
- Apply the symmetry relation correctly: $h[n] = h[M - 1 - n]$.

Q7.06.1 MCQ 2006 2M Ans: D ● ○ ○ CLICK FOR VIDEO SOLUTION

Theory: A digital filter rejects frequencies corresponding to its zeros lying on the unit circle.
Given,

$$H(z) = \frac{z^2 + 1}{z^2 + 0.81}.$$

The zeros are obtained from

$$z^2 + 1 = 0,$$

which gives

$$z = \pm j.$$

These zeros lie on the unit circle at

$$z = e^{\pm j\pi/2}.$$

Hence, the notch frequency of the filter is

$$\omega_0 = \frac{\pi}{2} \text{ rad/sample}.$$

The relation between analog frequency and digital frequency is

$$\omega = 2\pi \frac{f}{f_s}.$$

For rejecting a 50 Hz interference,

$$\frac{\pi}{2} = 2\pi \frac{50}{f_s}.$$

Therefore,

$$f_s = \frac{2\pi \times 50}{\pi/2} = 200 \text{ Hz}.$$

Hence,

$$\boxed{\text{(D) } 200 \text{ Hz}}$$

Key Points

- Zeros on the unit circle produce notch frequencies.
- Determine the notch frequency from the angle of the zeros.
- Use $\omega = 2\pi \frac{f}{f_s}$ to relate analog and digital frequencies.

Mistakes to be avoided

- Do not use the pole locations to determine the rejected frequency.
- Ensure the angle of the zero is measured in radians/sample.
- Convert between analog and digital frequencies using the sampling frequency.

Q7.05.1 | MCQ | 2005 | 2M | Ans: A | CLICK FOR VIDEO SOLUTION

Theory: The minimum order of a Butterworth filter is

$$N \geq \frac{\frac{1}{2}\log\left(\frac{\frac{1}{\delta_2^2}-1}{\frac{1}{\delta_1^2}-1}\right)}{\log\left(\frac{\Omega_s}{\Omega_p}\right)}.$$

From the specifications,

$$\delta_1 = 0.9, \qquad \delta_2 = 0.2,$$

$$\Omega_p = 2\pi \times 1000, \qquad \Omega_s = 2\pi \times 1500.$$

Hence,

$$\frac{\Omega_s}{\Omega_p} = \frac{1500}{1000} = 1.5.$$

Therefore,

$$N \geq \frac{\frac{1}{2}\log\left(\frac{\frac{1}{0.2^2}-1}{\frac{1}{0.9^2}-1}\right)}{\log(1.5)} = 5.70.$$

Since the filter order must be an integer,

$$N_{\min} = 6.$$

$$\boxed{\text{(A) } 6}$$

Key Points

- Use the Butterworth minimum-order formula.
- Compute the ratio Ω_s/Ω_p before substitution.
- Always round the order up to the next integer.

Mistakes to be avoided

- Do not round the computed order to the nearest integer.
- Use passband and stopband gains correctly.
- Do not forget that the common factor 2π cancels in the frequency ratio.

Q7.05.2

MCQ | 2005 | 2M | Ans: D | CLICK FOR VIDEO SOLUTION

Theory: Obtain the intermediate node equation and then express the output in terms of the input.
Let the intermediate node be

$$W(z).$$

From the signal-flow graph,

$$W(z) = X(z) + kz^{-1}W(z).$$

Hence,

$$W(z) = \frac{X(z)}{1 - kz^{-1}}.$$

The output is

$$Y(z) = -kX(z) + (1 - k^2)z^{-1}W(z).$$

Substituting the expression for $W(z)$,

$$Y(z) = -kX(z) + (1 - k^2)z^{-1}\frac{X(z)}{1 - kz^{-1}}.$$

Therefore,

$$\frac{Y(z)}{X(z)} = -k + \frac{(1 - k^2)z^{-1}}{1 - kz^{-1}}.$$

Taking the LCM,

$$H(z) = \frac{-k(1 - kz^{-1}) + (1 - k^2)z^{-1}}{1 - kz^{-1}}.$$

Simplifying,

$$H(z) = \frac{-k + z^{-1}}{1 - kz^{-1}}.$$

Hence,

$$\boxed{\text{(D) } \frac{-k + z^{-1}}{1 - kz^{-1}}}$$

Key Points

- Write the intermediate-node equation first.
- Solve for the intermediate variable before obtaining the transfer function.
- Express the final answer as $H(z) = Y(z)/X(z)$.

Mistakes to be avoided

- Do not miss the feedback term $kz^{-1}W(z)$.
- Combine fractions carefully while simplifying.
- Ensure the denominator corresponds to the feedback path.

Q7.05.3

MCQ | 2005 | 1M | Ans: D | ● ○ ○ | CLICK FOR VIDEO SOLUTION

Theory: The DTFT evaluated at $\omega = \pi$ is obtained by substituting $e^{-j\pi} = (-1)$.
The DTFT is

$$X(e^{j\omega}) = \sum_{n=0}^{6} x[n]e^{-j\omega n}.$$

At

$$\omega = \pi,$$
$$e^{-j\pi n} = (-1)^n.$$

Hence,

$$\begin{aligned} X(e^{j\pi}) &= 2(1) + 0(-1) - 1(1) - 3(-1) \\ &\quad + 4(1) + 1(-1) - 1(1) \\ &= 2 - 1 + 3 + 4 - 1 - 1 \\ &= 6. \end{aligned}$$

Therefore,

$$\boxed{\text{(D) } 6}$$

Key Points

- At $\omega = \pi$, $e^{-j\pi n} = (-1)^n$.
- Evaluate the DTFT directly using alternating signs.
- The sequence starts from $n = 0$.

Mistakes to be avoided

- Do not omit the zero-valued sample while assigning powers of (-1).
- Ensure the sequence indexing starts from $n = 0$.
- Do not introduce unnecessary factors of π while evaluating the DTFT.

Q7.04.1

MCQ | 2004 | 2M | Ans: A | ● ○ ○ | CLICK FOR VIDEO SOLUTION

Given

$$H(z) = \frac{1}{1 - 0.7z^{-1} + 0.13z^{-2}}.$$

From the given Direct Form realization,

$$C(z) = a_0\left[R(z) + a_1z^{-1}C(z) + a_2z^{-2}C(z)\right].$$

Therefore,

$$C(z)\left(1 - a_0a_1z^{-1} - a_0a_2z^{-2}\right) = a_0R(z).$$

Hence,

$$H(z) = \frac{C(z)}{R(z)} = \frac{a_0}{1 - a_0 a_1 z^{-1} - a_0 a_2 z^{-2}} = \frac{1}{\frac{1}{a_0} - a_1 z^{-1} - a_2 z^{-2}}.$$

Comparing with

$$H(z) = \frac{1}{1 - 0.7z^{-1} + 0.13z^{-2}},$$

we obtain

$$\frac{1}{a_0} = 1 \quad \Rightarrow \quad a_0 = 1,$$
$$a_1 = 0.7,$$
$$-a_2 = 0.13 \quad \Rightarrow \quad a_2 = -0.13.$$

Therefore,

$$\boxed{\text{(A) } 1.0,\ 0.7,\ -0.13}$$

Key Points

- Derive the transfer function directly from the realization before comparing coefficients.
- Compare both numerator and denominator carefully.
- The sign of the feedback coefficients is important.

Mistakes to be avoided

- Do not compare coefficients without first expressing the realization in transfer-function form.
- Do not forget that the denominator contains $-a_2 z^{-2}$.
- Verify the numerator to obtain a_0.

Q7.04.2 | MCQ | 2004 | 2M | Ans: 1.349 (MTA) | ● ○ ○ | CLICK FOR VIDEO SOLUTION

Given

$$f_T = 10\text{ kHz}, \qquad T = \frac{1}{f_T} = 10^{-4}\text{ s},$$
$$f_p = 2\text{ kHz}, \qquad f_s = 3\text{ kHz}.$$

The analog angular frequencies are

$$\Omega_p = 2\pi f_p = 4\pi \times 10^3,$$
$$\Omega_s = 2\pi f_s = 6\pi \times 10^3.$$

Using frequency pre-warping,

$$\bar{r} = 2\tan^{-1}\left(\frac{\Omega T}{2}\right).$$

Hence,

$$\bar{r}_p = 2\tan^{-1}(2\pi \times 0.1) = 2\tan^{-1}(0.2\pi) \approx 1.12,$$

and

$$\bar{r}_s = 2\tan^{-1}(3\pi \times 0.1) = 2\tan^{-1}(0.3\pi) \approx 1.511.$$

Therefore,

$$\frac{\bar{r}_s}{\bar{r}_p} = \frac{1.511}{1.12} \approx 1.349.$$

Since, none of the options match

$$\boxed{1.349\ (MTA)}$$

Key Points

- Pre-warping compensates for frequency distortion introduced by the bilinear transformation.
- Use $\bar{r} = 2\tan^{-1}(\Omega T/2)$.
- Compute the warped pass-band and stop-band frequencies separately before taking the ratio.

Mistakes to be avoided

- Do not use the unwarped ratio f_s/f_p.
- Convert frequency to angular frequency before applying the pre-warping formula.
- Ensure the sampling period is $T = 1/f_T$.

Q7.04.3 MCQ | 2004 | 2M | Ans: C | ● ○ ○ | CLICK FOR VIDEO SOLUTION

Theory: In the impulse invariance method, each analog pole s_i is mapped to the digital pole

$$z_i = e^{s_i T}.$$

Given,

$$H(s) = \frac{a}{s^2 + a^2}.$$

The poles are obtained from

$$s^2 + a^2 = 0,$$

which gives

$$s = \pm ja.$$

Applying impulse invariance,

$$z = e^{sT}.$$

Hence,

$$z_1 = e^{jaT}, \quad z_2 = e^{-jaT}.$$

Therefore, the poles of the equivalent digital filter are

$$\boxed{\text{(C) } e^{jaT},\ e^{-jaT}}$$

Key Points

- Impulse invariance maps poles using $z = e^{sT}$.
- Only the poles are mapped; zeros are not mapped directly.
- Imaginary-axis poles map onto the unit circle.

Mistakes to be avoided

- Do not use the bilinear transformation for impulse invariance questions.
- Do not substitute $s = \pm a$; the poles are $s = \pm ja$.
- Apply the exponential mapping to each pole individually.

Q7.04.4 | MCQ | 2004 | 2M | Ans: C | CLICK FOR VIDEO SOLUTION

Theory: For a first-order all-pass frequency transformation,

$$\tan\left(\frac{\omega'}{2}\right) = \frac{1-\alpha}{1+\alpha}\tan\left(\frac{\omega}{2}\right),$$

where

$$\omega = \frac{2\pi f_1}{f_s}, \qquad \omega' = \frac{2\pi f_2}{f_s}.$$

Given,

$$f_1 = 60\text{ Hz}, \qquad f_2 = 120\text{ Hz}, \qquad f_s = 400\text{ Hz}.$$

Hence,

$$\omega = 2\pi\frac{60}{400} = 0.3\pi,$$

and

$$\omega' = 2\pi\frac{120}{400} = 0.6\pi.$$

Substituting into the transformation equation,

$$\frac{1-\alpha}{1+\alpha} = \frac{\tan(0.3\pi)}{\tan(0.15\pi)} \approx 2.783.$$

Solving,

$$\alpha = \frac{1-2.783}{1+2.783} \approx -0.471.$$

Thus,

$$\alpha \approx -0.46.$$

Therefore,

$$\boxed{(c)\ \ -0.46}$$

Key Points

- Convert analog frequencies to normalized digital frequencies first.
- Use the LP-to-LP frequency transformation formula.
- Solve algebraically for α after substitution.

Mistakes to be avoided

- Do not substitute frequencies directly without normalization.
- Ensure the correct old and new notch frequencies are used.
- Keep the calculator in radian mode while evaluating the tangent function.

Q7.04.5 | MCQ | 2004 | 1M | Ans: C | CLICK FOR VIDEO SOLUTION

Theory: Multiplication by a complex exponential in the time domain corresponds to a frequency shift in the DTFT.
Given,

$$x[n] \xleftrightarrow{\text{DTFT}} X(e^{j\omega}).$$

Since

$$(-1)^n = e^{j\pi n},$$

we have

$$(-1)^n x[n] = e^{j\pi n} x[n].$$

Using the DTFT modulation property,

$$e^{j\omega_0 n} x[n] \xleftrightarrow{\text{DTFT}} X\left(e^{j(\omega-\omega_0)}\right),$$

with

$$\omega_0 = \pi,$$

we obtain

$$(-1)^n x[n] \xleftrightarrow{\text{DTFT}} X\left(e^{j(\omega-\pi)}\right).$$

Hence,

$$\boxed{\text{(C) } X\left(e^{j(\omega-\pi)}\right)}$$

Key Points

- $(-1)^n = e^{j\pi n}$.
- Multiplication by $e^{j\omega_0 n}$ shifts the DTFT by ω_0.
- DTFT modulation results in frequency translation.

Mistakes to be avoided

- Do not confuse modulation with differentiation.
- Remember the shift is $\omega - \omega_0$, not $\omega + \omega_0$.
- Express $(-1)^n$ as a complex exponential before applying the property.

Q7.03.1 MCQ | 2003 | 2M | Ans: C | ● ○ ○ | CLICK FOR VIDEO SOLUTION

Given

$$z_0 = \frac{1}{2} e^{j\pi/4}.$$

Since the FIR filter has a **real impulse response**,

$$z_0^* = \frac{1}{2} e^{-j\pi/4}$$

must also be a zero.
For a **linear-phase FIR filter**, every zero also has its reciprocal.
Hence,

$$\frac{1}{z_0} = 2e^{-j\pi/4}$$

is also a zero.
Taking the conjugate of the reciprocal gives

$$\left(\frac{1}{z_0}\right)^* = 2e^{j\pi/4}.$$

Therefore, the complete zero set is

$$\left\{\frac{1}{2}e^{j\pi/4}, \frac{1}{2}e^{-j\pi/4}, 2e^{-j\pi/4}, 2e^{j\pi/4}\right\}.$$

Among the given options, only Option (C) contains all the remaining zeros.

$$\boxed{\text{(C) } 2e^{-j\pi/4},\ 2e^{j\pi/4},\ \frac{1}{2}e^{-j\pi/4}}$$

Key Points

- A real FIR filter has zeros in complex conjugate pairs.
- A linear-phase FIR filter has reciprocal zero pairs.
- Combining both properties gives four related zeros.

Mistakes to be avoided

- Do not use only the conjugate property.
- Do not forget the reciprocal-conjugate zero.
- Reciprocal of $re^{j\theta}$ is $\frac{1}{r}e^{-j\theta}$.

Q7.03.2

MCQ | 2003 | 2M | Ans: D | CLICK FOR VIDEO SOLUTION

Given

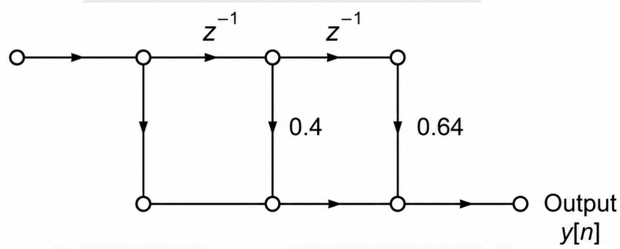

Fig. Solution Q7.03.2

The transfer function is

$$Y(z) = X(z) + 0.4z^{-1}X(z) + 0.64z^{-2}X(z).$$

Hence,

$$Y(z) = \left(1 + 0.4z^{-1} + 0.64z^{-2}\right) X(z).$$

From the lattice realization,

$$Y(z) = X(z) - K_2z^{-2}X(z) + K_1K_2z^{-1}X(z) - K_1z^{-1}X(z).$$

Therefore,

$$Y(z) = \left[1 + (K_1K_2 - K_1)z^{-1} - K_2z^{-2}\right] X(z).$$

Comparing coefficients,

$$-K_2 = 0.64$$

$$K_2 = -0.64.$$

Also,

$$K_1K_2 - K_1 = 0.4.$$

Substituting $K_2 = -0.64$,

$$-1.64K_1 = 0.4$$

$$K_1 = -0.244.$$

Hence,

$$K_1 = -0.244, \qquad K_2 = -0.640.$$

$$\boxed{\text{(D)} \; -0.244, \; -0.640}$$

Key Points

- First obtain the transfer function of the direct-form realization.
- Write the lattice transfer function in terms of K_1 and K_2.
- Compare coefficients of equal powers of z^{-1}.

Mistakes to be avoided

- Do not forget the negative sign before $K_2 z^{-2}$.
- Compare coefficients only after expressing both realizations in the same form.
- Solve for K_2 first before finding K_1.

Q7.03.3 MCQ 2003 2M Ans: D ● ○ ○ CLICK FOR VIDEO SOLUTION

Given

$$y[n] = x[n] - 2x[n-1] + x[n-2].$$

Taking the z-transform,

$$Y(z) = X(z) - 2z^{-1}X(z) + z^{-2}X(z).$$

Hence,

$$H(z) = \frac{Y(z)}{X(z)} = 1 - 2z^{-1} + z^{-2} = (1 - z^{-1})^2.$$

Evaluating on the unit circle,

$$H(e^{j\omega}) = 1 - 2e^{-j\omega} + e^{-j2\omega} = (1 - e^{-j\omega})^2.$$

Therefore,

$$|H(e^{j\omega})| = |1 - e^{-j\omega}|^2.$$

Now,

$$|1 - e^{-j\omega}| = \sqrt{(1-\cos\omega)^2 + \sin^2\omega} = \sqrt{2 - 2\cos\omega} = 2\sin\frac{\omega}{2}.$$

Hence,

$$|H(e^{j\omega})| = \left(2\sin\frac{\omega}{2}\right)^2.$$

Thus,

$$|H(e^{j\omega})| \to 0 \quad \text{for} \quad \omega \to 0,$$

and

$$|H(e^{j\omega})| \to 0 \quad \text{for} \quad \omega \to 2\pi.$$

Therefore, the filter attenuates low and very high frequencies while allowing a band of intermediate frequencies to pass.

$$\boxed{\text{D) — Band-pass filter passing } \frac{\pi}{8} \le |\omega| \le \frac{\pi}{4}}$$

Key Points

- Obtain the transfer function using the z-transform.
- Evaluate the frequency response on the unit circle.
- A magnitude response that is zero at both ends indicates a band-pass characteristic.

Mistakes to be avoided

- Do not stop after finding $H(z)$; always examine $H(e^{j\omega})$.
- Do not confuse $(1 - z^{-1})^2$ with a simple differentiator.
- Simplify the magnitude correctly using trigonometric identities.

Q7.03.4

MCQ | 2003 | 2M | Ans: B | CLICK FOR VIDEO SOLUTION

Concept In the impulse invariance method, aliasing occurs because the analog frequency response is periodically replicated after sampling.

To prevent aliasing, the input signal is first passed through an anti-aliasing low-pass filter so that it becomes band-limited before sampling.

Thus, the aliasing problem is avoided by pre-filtering the input signal.

Hence,

$$\boxed{\text{(B) — Pre-filtering the input signal to impose band-limitedness}}$$

Key Points

- Aliasing is caused by overlapping of replicated spectra after sampling.
- Anti-aliasing filtering removes high-frequency components before sampling.
- Bilinear transformation itself does not suffer from aliasing because it performs one-to-one frequency mapping.

Mistakes to be avoided

- Do not confuse aliasing with frequency warping.
- Mapping of the imaginary axis to the unit circle is a property of bilinear transformation, not the reason for eliminating aliasing.
- Up-sampling does not remove aliasing.

Q7.03.5

MCQ | 2003 | 2M | Ans: C | CLICK FOR VIDEO SOLUTION

Given

$$H(z) = 1 + z^{-1} + z^{-2}.$$

The frequency response is

$$H(e^{j\Omega}) = 1 + e^{-j\Omega} + e^{-j2\Omega}.$$

Factorizing,

$$H(e^{j\Omega}) = e^{-j\Omega}\,(1 + 2\cos\Omega)\,.$$

For complete rejection,

$$H(e^{j\Omega}) = 0.$$

Hence,

$$1 + 2\cos\Omega = 0,$$

or,

$$\cos\Omega = -\frac{1}{2}.$$

Therefore,

$$\Omega = \frac{2\pi}{3}.$$

Using

$$\Omega = 2\pi\frac{f}{f_s},$$

with

$$f_s = 18 \text{ kHz},$$

we obtain

$$f = \frac{\Omega}{2\pi}f_s = \frac{2\pi/3}{2\pi} \times 18 = 6 \text{ kHz}.$$

Hence,

$$\boxed{\text{(B) 6 kHz}}$$

Key Points

- The frequencies rejected by the filter correspond to the zeros of $H(e^{j\Omega})$.
- Use the relation $\Omega = 2\pi f/f_s$ to convert digital frequency to analog frequency.
- Factorization simplifies finding the frequency response zeros.

Mistakes to be avoided

- Do not substitute analog frequency directly into $H(z)$.
- Use the sampling frequency while converting between analog and digital frequencies.
- Do not ignore the periodicity of the digital frequency response.

Q7.03.5 MCQ | 2003 | 2M | Ans: C | CLICK FOR VIDEO SOLUTION

Given

$$\omega = \frac{z-1}{z+1}.$$

For a point inside the unit circle,

$$|z| < 1.$$

Consider the real interval

$$0 < z < 1.$$

Then,

$$z - 1 < 0, \qquad z + 1 > 0.$$

Hence,

$$\omega = \frac{z-1}{z+1} < 0.$$

Therefore, points inside the unit circle are mapped to the left half of the ω-plane.
Thus,

$$\boxed{\text{(A) Maps the inside of the unit circle in the } z\text{-plane to the left half of the } \omega\text{-plane}}$$

Key Points

- The bilinear transformation maps the unit circle in the z-plane onto the imaginary axis of the transformed plane.
- The inside of the unit circle maps to the left half-plane.

- Stability is preserved under bilinear transformation.

Mistakes to be avoided

- Do not confuse the mapping direction of the inside and outside of the unit circle.
- Remember that the transformation preserves stability.
- Do not interchange the left-half plane with the right-half plane.

Q7.96.1 MCQ 1996 2M Ans: A ● ○ ○ CLICK FOR VIDEO SOLUTION

Given A linear phase low-pass filter shifts a 50 Hz signal by 3 ms. A linear phase filter has a constant group delay over its passband.

Hence, every frequency component within the passband experiences the same time delay.

Since both 50 Hz and 150 Hz lie within the passband (f_c = 1500 Hz),

$$\boxed{\text{Delay at 150 Hz} = 3\text{ ms.}}$$

Therefore,

$$\boxed{\text{(A) 3 ms}}$$

Key Points

- A linear phase filter provides constant group delay.
- All frequency components within the passband experience the same time delay.
- Group delay is independent of frequency for an ideal linear phase filter.

Mistakes to be avoided

- Do not assume the delay is proportional to frequency.
- Do not confuse phase delay with group delay.

DEBUG INFO

Chapter VII

DTFT, DTFS, DFT & Miscellaneous

Total Questions : 35

MCQ : 30

MSQ : 0

NAT : 5

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